7+1: provenance

Sources, derivations, corrections and the verification record of the notebook 7+1.nb and of its documentation

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This record states, for every object of 7+1.nb, where it comes from (the task statement, the notebook 4+4.nb, or the literature), the equation that defines it, how it was derived, and how it was checked: the full list of the 346 checks and their results, the independent Python cross-checks, and the hashes of the files. Sections 1–3 concern the first stage (sections 0–6 of the notebook: octonions and Clifford algebras), section 6 the second stage (sections 7–10: the Dirac equation and the Kohn–Sham theories of the field dirac16complex), section 7 the third stage (sections 11–20: gravity, the self-coupled field in gravitational fields, the Standard Model).

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1. Sources

1.1 The task statement

The task asked for a new Mathematica notebook 7+1.nb in `D:\Developer\github\7+1` defining a new realization of the octonions and a new representation of the Clifford algebra of an 8-dimensional vector space that mirror, as far as possible, the split octonions and the pseudo-Clifford algebra of 4+4.nb; real numbers and the real 8×8 and 16×16 matrices of 4+4.nb were to be used if possible, otherwise their complexification; examples were to check all work; notation was to be changed and documented where confusing. It fixed the coordinates

$$\begin{aligned}
&\text{4+4.nb:}\quad X=(x_0,\dots,x_7),\ x_0=\text{hidden space-like},\ x_1,x_2,x_3=\text{3-space},\\
&\qquad\qquad x_4=\text{time},\ x_5,x_6,x_7=\text{auxiliary time-like},\\
&\text{7+1.nb:}\quad X=(x_0,\dots,x_7),\ x_0=\text{time (octonion unit)},\ x_1,x_2,x_3=\text{3-space},\\
&\qquad\qquad x_4,\dots,x_7=\text{auxiliary space-like},
\end{aligned}$$

and gave two definitions. [1A] the octonions: $e_0=1$, $e_i^2=-1$, Fano plane, Cayley–Dickson doubling $\mathbb{O}=\mathbb{H}\oplus\mathbb{H}$, conjugation $\bar{x}=x_0-\sum_i x_ie_i$ with $\overline{xy}=\bar{y}\bar{x}$, $N(x)=x\bar{x}=\sum_A x_A^2$, $N(xy)=N(x)N(y)$, $x^{-1}=\bar{x}/N(x)$, alternativity, power-associativity. [1B] the Clifford algebra $Cl(V,Q)=\mathcal{T}(V)/\mathcal{I}_Q(V)$ with $v^2=Q(v)1$, $vw+wv=2B(v,w)1$, $B(v,w)=\frac{1}{2}(Q(v+w)-Q(v)-Q(w))$, dimension $2^8=256$, a classification table for $p+q=8$, and $Cl_8(\mathbb{C})\cong\mathbb{C}(16)$. Section 4 lists the statements of [1A] and [1B] that had to be corrected.

1.2 The notebook 4+4.nb

File `4+4.nb`, 9348656 bytes, written by Wolfram 15.0 (header `CreatedBy='Wolfram 15.0'`), SHA-256 `740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5`. It was read with the parser `tools/nbparse.py` (806 cells: 366 input, 366 output, 49 text). Every object used by 7+1.nb was reproduced with the formula of the cell given here, and each reproduction is checked in 7+1.nb section 2:

object	4+4.nb cell	definition in 4+4.nb
$\eta^{(4,4)}$	In[9]	$\mathrm{diag}(1_4,-1_4)$ (`η4488`)
σ	In[18]	$\left(\begin{smallmatrix}0 & 1_4\\ 1_4 & 0\end{smallmatrix}\right)$ (`σ=σ88`)
s_h,t_h	In[29], In[35], In[36]	$(s_h)_{pq}=\epsilon_{hpq4}-(\delta_{p4}\delta_{qh}-\delta_{ph}\delta_{q4})$, $(t_h)_{pq}=\epsilon_{hpq4}+(\dots)$
ζ_A	In[60]–In[63]	$\zeta_0=1_8$, $\zeta_{7-h}=\left(\begin{smallmatrix}0 & t_h\\ -t_h & 0\end{smallmatrix}\right)$, $\zeta_h=\left(\begin{smallmatrix}0 & s_h\\ s_h & 0\end{smallmatrix}\right)$, $\zeta_7=\zeta_1\cdots\zeta_6$
relations of ζ	In[64], In[65], In[67]	$\sigma=\zeta_1\zeta_2\zeta_3$; $\frac{1}{2}\{\zeta_A,\zeta_B\}=-\eta^{(4,4)}_{AB}1_8$; $-\eta^{(4,4)}_{AA}\zeta_A=\zeta_A^{\mathsf{T}}$
τ_A	In[70]	$\tau_0=1_8$, $\tau_A=\zeta_A$
$\bar{\tau}_A$	In[74]	$\bar{\tau}_0=1_8$, $\bar{\tau}_A=\sigma\tau_A^{\mathsf{T}}\sigma$ (symbolized tau-bar)
relations of τ	In[77], In[87], In[89]	$\frac{1}{2}(\tau_A\bar{\tau}_B+\tau_B\bar{\tau}_A)=\eta^{(4,4)}_{AB}1_8$; $\tau_1\cdots\tau_7=1_8$; $\sigma=\tau_4\tau_5\tau_6\tau_7$
T_A	In[98]	$T_A=\left(\begin{smallmatrix}0 & \bar{\tau}_A\\ \tau_A & 0\end{smallmatrix}\right)$ (`T16^A[A1]`)
T_8, σ_{16}	In[99], In[97], In[106]	$T_8=T_0\cdots T_7$; $\sigma_{16}=T_0T_1T_2T_3=\mathrm{diag}(-\sigma,\sigma)$
Clifford relation	In[110]	$\frac{1}{2}\{T_A,T_B\}=\eta^{(4,4)}_{AB}1_{16}$
P_L,P_R	In[112]–In[114]	$P_L=\frac{1}{2}(1_{16}-T_8)$, $P_R=\frac{1}{2}(1_{16}+T_8)$
U_h	In[257], In[259]	eigenvectors of σ divided by $\sqrt{2}$
u	In[264], In[266]	`hUSE = 8`, $u=U_8=(1,0,0,0,1,0,0,0)/\sqrt{2}$
g	In[283]	$g_{AB}=\eta^{(4,4)}$
E_{44}, E_{44}^{-1}	In[294]–In[296]	$(E_{44})_{Ba}=(U_h^{\mathsf{T}}\sigma\bar{\tau}_B)_a$; $E_{44}^{-1}=(U_h^{\mathsf{T}}\sigma U_h)(\eta^{(4,4)}_{BB}\tau_BU_h)_B$
ε_B	In[313]	`EA = Array[eA, 8]`
split table	In[325], Out[325]	$\varepsilon_B\varepsilon_C=\sum_D(g_{BB}E_{44}\tau_{B-1}E_{44}^{-1})_{DC}\varepsilon_D$ (`gridx`)

1.3 Literature

- J. C. Baez, *The Octonions*, Bull. Amer. Math. Soc. 39 (2002) 145–205, arXiv:math/0105155, section 2.2 (the page https://math.ucr.edu/home/baez/octonions/node5.html, which the task statement itself cites, was read on 2026-10-27): the Cayley–Dickson product $(a,b)(c,d) = (ac - db^*,\ a^*d + cb)$, where Baez writes x^* for the conjugate $\bar{x}$ and the conjugate of a pair is $(a,b)^* = (a^*, -b)$. 7+1.nb proves that its octonions satisfy this product with $x = a + \ell b$, $\ell = e_4$.
- H. B. Lawson and M.-L. Michelsohn, *Spin Geometry*, Princeton (1989): classification of real Clifford algebras, $Cl(p+1,q+1) \cong Cl(p,q) \otimes \mathbb{R}(2)$. 7+1.nb does not rely on it: it computes the classification directly (section 2.10 below).
- Standard facts used in proofs: Schur's lemma (the intertwiners of an irreducible module form a division algebra; over $\mathbb{C}$, the scalars); $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$ has a unique irreducible module $\mathbb{C}^{16}$; Hurwitz's theorem (a unital composition algebra of dimension 8 is the octonions if its norm is positive definite and the split octonions if the norm is indefinite); Degen's eight-square identity.

2. Derivations: how each new object was obtained

2.1 Octonion matrices τ^O

Origin: the split matrices of 4+4.nb and the observation (checked) that $\sigma = \tau_1\tau_2\tau_3 = \tau_4\tau_5\tau_6\tau_7$ commutes with τ_1, τ_2, τ_3, anticommutes with $\tau_4, \ldots, \tau_7$ and squares to 1_8. Hence

$$\tau_0^O = 1_8, \quad \tau_A^O = \tau_A \ (A \le 3), \quad \tau_k^O = \tau_k\sigma \ (k \ge 4), \qquad (\tau_k\sigma)^2 = \tau_k\sigma\tau_k\sigma = -\tau_k^2\sigma^2 = -1_8,$$
$$\tau_h^O\tau_k^O = \tau_h\tau_k\sigma = -\tau_k\tau_h\sigma = -\tau_k\sigma\tau_h = -\tau_k^O\tau_h^O, \qquad \tau_k^O\tau_l^O = \tau_k\sigma\tau_l\sigma = -\tau_k\tau_l = \tau_l\tau_k = -\tau_l^O\tau_k^O \quad (k \ne l \ge 4).$$

This is the real counterpart of the complex substitution $\tau_k \to i\tau_k$ that turns split octonions into octonions over $\mathbb{C}$.

2.2 Octonion vielbein and multiplication constants

Origin: In[294] and In[325] of 4+4.nb with the split spinor metric σ replaced by the Euclidean $g^O = 1_8$ and the same unit spinor u:

$$(E^O)_{Ba} = \left(u^{\mathsf{T}} g^O \bar{\tau}_B^O\right)_a, \qquad L_A^O = g_{AA}^O E^O \tau_A^O (E^O)^{-1}, \qquad e_A e_K = \sum_L (L_A^O)_{LK} e_L.$$

Why the result is the octonion algebra: $(E^O)^{-1}e_0 = u$ makes e_0 a two-sided unit, and $\tau^O(x)\bar{\tau}^O(x) = N(x)1_8$ together with the orthogonality of E^O gives $N(xy) = N(x)N(y)$. A unital composition algebra of dimension 8 with a positive definite norm is the octonions (Hurwitz). The notebook checks all the laws directly.

2.3 Cayley–Dickson form

Origin: Baez 2002, section 2.2. With $\ell = e_4$, $x = a + \ell b$, $a = (x_0, \ldots, x_3)$, $b = (x_4, \ldots, x_7)$:

$$(a,b)(c,d) = \left(ac + \gamma\, d\bar{b},\ \bar{a}d + cb\right), \qquad \gamma = -1 \ (7{+}1.\text{nb}), \qquad \gamma = +1 \ (4{+}4.\text{nb}).$$

2.4 Euclidean mirror T^O

Origin: In[98] of 4+4.nb with $\tau \to \tau^O$: $T_A^O = \left(\begin{smallmatrix} 0 & \bar{\tau}_A^O \\ \tau_A^O & 0 \end{smallmatrix}\right)$, so that $\{T_A^O, T_B^O\} = \left(\begin{smallmatrix} \bar{\tau}_A^O\tau_B^O + \bar{\tau}_B^O\tau_A^O & 0 \\ 0 & \tau_A^O\bar{\tau}_B^O + \tau_B^O\bar{\tau}_A^O \end{smallmatrix}\right) = 2\delta_{AB}1_{16}$. In terms of 4+4.nb: $T_A^O = T_A$ $(A \le 3)$, $T_k^O = T_k S_{44}$ $(k \ge 4)$, $S_{44} = T_4T_5T_6T_7 = \mathrm{diag}(\sigma, \sigma)$.

2.5 Real 7+1 generators $\Gamma^{(1,7)}$

Origin: the standard construction of a Lorentzian Clifford algebra from a Euclidean one, applied to T^O:

$$\Gamma_0^{(1,7)} = T_0^O, \qquad \Gamma_i^{(1,7)} = T_0^O T_i^O, \qquad (T_0^O T_i^O)^2 = -(T_0^O)^2 (T_i^O)^2 = -1_{16},$$
$$\{T_0^O, T_0^O T_i^O\} = T_i^O + T_0^O T_i^O T_0^O = T_i^O - T_i^O = 0, \qquad \{T_0^O T_i^O, T_0^O T_j^O\} = -\{T_i^O, T_j^O\} = -2\delta_{ij} 1_{16}.$$

Hence $\{\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}\} = 2\eta_{AB}^{(1,7)} 1_{16}$, and $\Gamma^{(1,7)}(x)^2 = \mathrm{Re}(xx) 1_{16}$.

2.6 Complexified generators $\Gamma^{(7,1)}$

Origin: the task's instruction to use the complexification of the 4+4 matrices if real numbers fail. Only the time block is multiplied by i: $\tau_0^C = \bar\tau_0^C = i 1_8$, $\tau_k^C = \tau_k^O$; then $\Gamma_A^{(7,1)} = \left(\begin{smallmatrix} 0 & \bar\tau_A^C \\ \tau_A^C & 0 \end{smallmatrix}\right)$, $(\Gamma_0^{(7,1)})^2 = i^2\, 1_{16} = -1_{16}$, and $\Gamma_A^{(7,1)} = S(i\Gamma_A^{(1,7)})S^{-1}$ with $S = (1_{16} - i\Gamma_0^{(1,7)})/\sqrt{2}$.

2.7 The impossibility theorem

Origin: Schur's lemma and the uniqueness of the irreducible module of $Cl_8(\mathbb{C}) \cong \mathbb{C}(16)$. The intertwiners M with $M\overline{\Gamma_A^{(7,1)}} = \Gamma_A^{(7,1)} M$ are computed in 7+1.nb with `NullSpace`; they form the line through $M_0 = \Gamma_8^{(1,7)} = \left(\begin{smallmatrix} 0 & -1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, and $M_0\overline{M_0} = -1_{16}$. A real realization $R_A = T\Gamma_A^{(7,1)}T^{-1}$ would give an intertwiner $M = T^{-1}\overline{T}$ with $M\overline{M} = 1_{16}$, impossible because $M = cM_0$ gives $M\overline{M} = -|c|^2 1_{16}$.

2.8 Real 32×32 generators

Origin: realification. $\widehat\Gamma_A^{(7,1)} = \varepsilon \otimes \Gamma_A^{(1,7)}$ with $\varepsilon^2 = -1_2$ gives $\{\widehat\Gamma_A^{(7,1)}, \widehat\Gamma_B^{(7,1)}\} = \varepsilon^2 \otimes 2\eta_{AB}^{(1,7)} 1_{16} = 2\eta_{AB}^{(7,1)} 1_{32}$. Commutant: $Q_i = \varepsilon \otimes 1_{16}$, $Q_j = \sigma_1 \otimes \Gamma_8^{(1,7)}$, $Q_k = \sigma_3 \otimes \Gamma_8^{(1,7)}$, found from $\varepsilon\sigma_1 = -\sigma_1\varepsilon$, $\varepsilon\sigma_3 = -\sigma_3\varepsilon$ and $\Gamma_8^{(1,7)}\Gamma_A^{(1,7)} = -\Gamma_A^{(1,7)}\Gamma_8^{(1,7)}$.

2.9 Examples

so(1,7): $\Sigma_{AB} = \frac{1}{4}[\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}]$; the commutation relations follow from the Clifford relation. Boost and rotation: $(\Gamma_0^{(1,7)}\Gamma_1^{(1,7)})^2 = 1_{16}$, $(\Gamma_4^{(1,7)}\Gamma_5^{(1,7)})^2 = -1_{16}$, so $e^{\beta\Sigma_{01}} = \cosh\frac{\beta}{2} + \sinh\frac{\beta}{2}\Gamma_0^{(1,7)}\Gamma_1^{(1,7)}$ and $e^{\theta\Sigma_{45}} = \cos\frac{\theta}{2} + \sin\frac{\theta}{2}\Gamma_4^{(1,7)}\Gamma_5^{(1,7)}$. Dirac operator: $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A, \Gamma^B\}\partial_A\partial_B = \eta_{(1,7)}^{AB}\partial_A\partial_B$. Real amplitudes: $p = (3,2,2,0,0,0,0,0)$, $p^{\mathsf T}\eta^{(1,7)}p = 1$, $\not{p}^{\,2} = 1_{16}$; the real matrix $\not{p}$ has a real 8-dimensional eigenspace for the eigenvalue $+1$.

Real spinor fields. If J anticommutes with every $\Gamma_A^{(1,7)}$, then $J\Gamma_8^{(1,7)}$ commutes with every $\Gamma_A^{(1,7)}$ (because $\Gamma_8^{(1,7)}$ anticommutes with each of them), so by irreducibility $J\Gamma_8^{(1,7)} = c\, 1_{16}$ and $J = -c\,\Gamma_8^{(1,7)}$ (as $(\Gamma_8^{(1,7)})^{-1} = -\Gamma_8^{(1,7)}$), whose square is $-c^2\, 1_{16}$. Hence the real operator $\Gamma^A\partial_A - mJ$ has the square $\partial_0^2 - \cdots - \partial_7^2 - c^2m^2$ (a tachyon), and a real equation $(G^A\partial_A - m)\Psi = 0$ implies the Klein–Gordon equation only if the real G_A obey the $\eta^{(7,1)}$ relation, impossible for 16×16 matrices (2.7). Massless: for $k = (1,1,0,\ldots,0)$, $\not{k}^{\,2} = k^{\mathsf T}\eta^{(1,7)}k\, 1_{16} = 0$ and $\Gamma^A\partial_A[w\cos(k\cdot x)] = -\sin(k\cdot x)\ \not{k}\, w$, which vanishes for the 8 real solutions of $\not{k}\, w = 0$. Massive, with the real 32×32 matrices: $P = \sum_A p^A\widehat\Gamma_A^{(7,1)}$, $P^2 = -m^2\, 1_{32}$, and with $\phi = p\cdot x$, $\widehat\Gamma^A = \eta_{(7,1)}^{AA}\widehat\Gamma_A^{(7,1)}$:

$$\begin{aligned}\widehat\Gamma^A\partial_A\Big(w\cos\phi + \tfrac{1}{m}Pw\sin\phi\Big) &= -P\Big(-w\sin\phi + \tfrac{1}{m}Pw\cos\phi\Big)\\ &= Pw\sin\phi + m\,w\cos\phi = m\Big(w\cos\phi + \tfrac{1}{m}Pw\sin\phi\Big),\end{aligned}$$

using $\partial_A\phi = (\eta^{(1,7)}p)_A$ and $\eta_{(7,1)}^{AA}\eta_{AA}^{(1,7)} = -1$.

2.10 Classification of the nine signatures

For each p, complex generators $G_A = T_A^O$ $(A < p)$, $G_A = iT_A^O$ $(A \geq p)$ satisfy $\{G_A, G_B\} = \pm 2\delta_{AB} 1_{16}$ with p plus signs. The indicator sign of $M\overline{M}$ ($M\overline{G_A} = G_A M$) is $+1$ for $p \in \{8,5,4,1,0\}$ ($\mathbb{R}(16)$) and -1 for $p \in \{7,6,3,2\}$ ($\mathbb{H}(8)$). An independent numerical method (`tools/classify_clifford.py`: commutant of the realified 32-dimensional representation, whose traceless part has a negative definite trace form exactly in the quaternionic case) gives the same table. Explicit real generators: T^O (8,0), $\Gamma_0^{(5,3)} = T_0$, $\Gamma_i^{(5,3)} = T_0 T_i$ (5,3) with $\eta^{(5,3)} = \mathrm{diag}(1,-1,-1,-1,1,1,1,1)$, T (4,4), $\Gamma^{(1,7)}$ (1,7), $\Gamma_A^{(0,8)} = T_8^O T_A^O$ (0,8).

3. Corrections to the task statement (with the evidence in 7+1.nb)

statement in the task	**correction**	**evidence (check in 7+1.nb)**
[1A]: $(a,b)(c,d) = (ac - \bar{d}b,\ \bar{a}d + cb)$ defines the octonions	not alternative, norm not multiplicative; correct form $(ac - d\bar{b},\ \bar{a}d + cb)$ (Baez 2002, 2.2)	"the formula ... of [1A] is NOT alternative", "... its norm is not multiplicative", "Cayley-Dickson, γ = -1 ..."
[1A]: first formula $(ac - \bar{d}b,\ da + cb\bar{x})$	contains a stray "x"; not interpretable	—
[1B]: $1+8+28+56+72+56+28+8+1 = 256$	$\binom{8}{4} = 70$	"grades: binomial(8,k) == ..."
[1B]: $Cl(7,1) \cong \mathbb{R}(16)$	$Cl(7,1) \cong \mathbb{H}(8)$ for $v^2 = Q(v)$	"indicator(Γ71) == -1 ...", "classification: ..."
[1B]: $Cl(5,3) \cong \mathbb{C}(16)$, $Cl(3,5) \cong \mathbb{C}(16)$	$Cl(5,3) \cong \mathbb{R}(16)$, $Cl(3,5) \cong \mathbb{H}(8)$; $\mathbb{C}(16)$ has real dimension 512	"classification: ...", "$\mathbb{C}(16)$ cannot occur ..."
[1B]: metric-sign strings for (6,2), (5,3), (4,4), (2,6)	mistyped (wrong number of signs)	table of section 5.6 of 7+1.nb
split octonions are a "pseudo-normed division algebra"	they have zero divisors: $(\varepsilon_1 + \varepsilon_5)(\varepsilon_1 - \varepsilon_5) = 0$	"split: (eA[1] + eA[5])(eA[1] - eA[5]) == 0 ..."

4. Verification record

4.1 The Mathematica run

item	**value**
program	`C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe`
$Version	15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)
method	`Get["D:/Developer/github/7+1/tools/run_eval.wl"]`: all cells evaluated (`SelectionEvaluate`) in a fresh kernel (front-end kernel configuration "Eval71"), then `NotebookSave`; earlier stages: `NotebookEvaluate[nb, InsertResults -> True]` in the front end's kernel
finished	Tue 27 Oct 2026 14:15:35
evaluation time	225.3 s
private context of the notebook	`Notebook$$41$124309`
input cells / output cells / message cells	145 / 382 / 0
checks / True	346 / 346 (310 printed where they stand, 36 by `frameChecks` in the summary table)
commutant computation (8192×1024 sparse `NullSpace`)	0.9989724 s
SHA-256 of the evaluated 7+1.nb (1034401 bytes)	`060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a`

4.2 All checks of 7+1.nb and their results

Each row is one check `check["name", test]` of the notebook, in the order in which the notebook makes them (the order of the summary table of its last section), grouped by the section of the notebook in which it stands; the test is the Wolfram Language expression of that check, and the result is what Wolfram 15.0.1 returned. The 36 checks of the generalized frame matrices are made by the function `frameChecks` of section 13.1, nine for each of four metrics, in sections 13.2 (cosmological metric), 13.3 (pp-wave, inhomogeneous metric) and 13.5 (static radial metric), where they are listed; only the summary table of section 20 prints them.

2.1 σ, the self-dual and anti-self-dual blocks, ζ, τ and τbar

#	check	result
1	4+4: ζ[7] == diag(-1,-1,-1,-1,1,1,1,1)	True
2	4+4: σ == ζ[1].ζ[2].ζ[3]	True
3	4+4: (ζ[A].ζ[B] + ζ[B].ζ[A])/2 == -η4488[[A+1,B+1]] ID8 (A,B = 1..7)	True
4	4+4: -η4488[[A+1,A+1]] ζ[A] == Transpose[ζ[A]] (A = 1..7)	True
5	4+4: (τ[A].τbar[B] + τ[B].τbar[A])/2 == η4488[[A+1,B+1]] ID8	True
6	4+4: (τbar[A].τ[B] + τbar[B].τ[A])/2 == η4488[[A+1,B+1]] ID8	True
7	4+4: τ[1].τ[2]...τ[7] == ID8	True
8	4+4: σ == τ[1].τ[2].τ[3] == τ[4].τ[5].τ[6].τ[7]	True
9	4+4: σ commutes with τ[1..3] and anticommutes with τ[4..7]; σ.σ == ID8	True

2.2 The 4+4 Dirac matrices T16 and the chiral projectors

#	check	result
10	4+4: (T16[A].T16[B] + T16[B].T16[A])/2 == η4488[[A+1,B+1]] ID16	True
11	4+4: T16[A] real (entries 0, ±1)	True
12	4+4: T16[8] == diag(-ID8, ID8)	True
13	4+4: σ16 == diag(-σ, σ)	True
14	4+4: PL44 + PR44 == ID16, PL44.PL44 == PL44, PR44.PR44 == PR44, PL44.PR44 == 0	True

2.3 The split-octonion multiplication constants (gridx of 4+4.nb) and how they come from τ

#	check	result
15	4+4: uBasis == evecs/Sqrt[2] from Eigensystem[σ] (the basis of 4+4.nb)	True
16	4+4: unit == (1,0,0,0,1,0,0,0)/Sqrt[2], σ.unit == unit, unit.unit == 1, unit.σ.unit == 1	True
17	4+4 (In[294]): E44inv[h] == (u.σ.u) Transpose[Table[η4488[[B,B]] τ[B-1].u, {B,1,8}]] for h = 1..8	True
18	4+4: gridx44 is identical to the split-octonion table printed by 4+4.nb	True
19	4+4: LS[A] are signed permutation matrices (constants 0, ±1)	True

2.4 Split octonions: composition algebra, but not a division algebra

#	check	result
20	split: eA[1] is the identity; eA[2..4]^2 == -eA[1]; eA[5..8]^2 == +eA[1]	True
21	split: x x̄ == N(x) eA[1] with N = η4488 (symbolic)	True
22	split: N(xy) == N(x) N(y) (symbolic)	True
23	split: alternative x(xy) == (xx)y and (yx)x == y(xx) (symbolic)	True
24	split: (eA[1] + eA[5])(eA[1] - eA[5]) == 0 with both factors nonzero: NOT a division algebra	True

3.1 From the split matrices τ to the octonionic matrices τO

#	check	result
25	τO[A] (A = 1..7): real, antisymmetric, τO[A].τO[A] == -ID8	True
26	τO[A] anticommute: τO[A].τO[B] == -τO[B].τO[A] (A != B, A,B = 1..7)	True
27	mirror of 4+4: (τO[A].τObar[B] + τO[B].τObar[A])/2 == gO[[A+1,B+1]] ID8	True
28	mirror of 4+4: (τObar[A].τO[B] + τObar[B].τO[A])/2 == gO[[A+1,B+1]] ID8	True
29	mirror of 4+4: τO[1].τO[2]...τO[7] == ID8	True
30	mirror of 4+4: τObar[A] == gO.Transpose[τO[A]].gO, as τbar[A] == σ.Transpose[τ[A]].σ	True
31	τO[4..7] are products of three split τ: τO[4] == τ[5].τ[6].τ[7], τO[5] == -τ[4].τ[6].τ[7], τO[6] == τ[4].τ[5].τ[7], τO[7] == -τ[4].τ[5].τ[6]	True

3.2 The octonion vielbein (the same unit spinor u as in 4+4.nb)

#	check	result
32	σ.unit == unit, hence τO[A].unit == τ[A].unit (A = 0..7)	True
33	EO is orthogonal: EOinv == Transpose[EO]	True
34	mirror of 4+4.nb In[294]: EOinv == (u.gO.u) Transpose[Table[gO[[B,B]] τO[B-1].u, {B,1,8}]]	True
35	EO == η4488.E44[hUSE]: the two vielbeins differ only by the signs of rows 5..8	True

3.3 The octonion multiplication constants: gridO, the mirror of gridx

#	check	result
36	gridO equals the octonion table computed independently in Python	True
37	LO[A] are signed permutation matrices (all multiplication constants are 0, ±1)	True
38	oMul reproduces gridO on all 64 pairs of basis units	True

3.4 The properties listed in [1A], verified

#	check	result
39	e[0] is a two-sided identity	True
40	e[i] e[i] == -e[0] (i = 1..7)	True
41	e[i] e[j] == -e[j] e[i] (i != j, i,j = 1..7): non-commutative	True
42	7 Fano lines; every pair of distinct imaginary units lies on exactly one line	True
43	each oriented line (a,b,c) is cyclic: e[a]e[b] == e[c], e[b]e[c] == e[a], e[c]e[a] == e[b]	True
44	each line spans a quaternion subalgebra: {e[0], e[a], e[b], e[c]} closed and associative	True
45	the picture's 7 lines are exactly fanoLines	True
46	left alternative: x(xy) == (xx)y (symbolic)	True
47	right alternative: (yx)x == y(xx) (symbolic)	True
48	flexible: (xy)x == x(yx) (symbolic)	True
49	associator is alternating: [x,y,z] == -[y,x,z] == -[x,z,y] (symbolic)	True
50	power associativity: x(xx) == (xx)x (symbolic)	True
51	Moufang: z(x(zy)) == ((zx)z)y, x(z(yz)) == ((xz)y)z, (zx)(yz) == (z(xy))z (x, y basis units, z symbolic)	True
52	x x̄ == x̄ x == N(x) e[0] (symbolic)	True
53	conjugation is an anti-automorphism: conj(xy) == conj(y) conj(x) (symbolic)	True
54	Re(x) == (x + x̄)/2 == x0	True
55	norm is multiplicative: N(xy) == N(x) N(y) (symbolic; Degen's eight-square identity)	True
56	inverse x^-1 = x̄/N(x): x x^-1 == x^-1 x == e[0] (symbolic)	True
57	every x satisfies x x - 2 Re(x) x + N(x) e[0] == 0, so the subalgebra generated by x is $\mathbb{R}$ or $\mathbb{C}$	True
58	exactly 168 of the 512 ordered basis triples are non-associative (= 7*6*4: distinct imaginary units not on one Fano line)	True

#	check	result
59	example: (e[1]e[2])e[4] == e[7] but e[1](e[2]e[4]) == -e[7]	True
60	numerical example: N(x y) == N(x) N(y) == 15*6 == 90 and x x^-1 == e[0]	True
61	numerical example: x y != y x and (x y) z != x (y z)	True

3.5 Cayley–Dickson doubling: $\mathbb{O} = \mathbb{H} \oplus \ell\mathbb{H}$

#	check	result
62	$\mathbb{H}$ = span{e[0],e[1],e[2],e[3]} is a subalgebra	True
63	ℓ = e[4] doubles $\mathbb{H}$: e[4] e[k] == e[4+k] (k = 1,2,3) and e[4] e[4] == -e[0]	True
64	orientation inherited from 4+4.nb: e[1] e[2] == -e[3]	True
65	Cayley-Dickson, γ = -1: (ac - d $\bar{b}$, $\bar{a}$ d + c b) reproduces the octonion product (symbolic)	True
66	Cayley-Dickson, γ = +1: (ac + d $\bar{b}$, $\bar{a}$ d + c b) reproduces the SPLIT product of 4+4.nb (symbolic)	True
67	the formula (ac - $\bar{d}$ b, $\bar{a}$ d + c b) of [1A] is NOT alternative: x·x·y != x(xy) for x = e[0]+e[4], y = e[1]	True
68	... and its norm is not multiplicative (symbolic)	True

3.6 Octonions versus split octonions

#	check	result
69	e[A]e[B] (octonion) == s_AB e[A]e[B] (split) for all A, B = 0..7	True
70	over $\mathbb{C}$: ϕ(x *split y) == ϕ(x) *oct ϕ(y) (symbolic): split $\otimes$ $\mathbb{C}$ $\cong$ $\mathbb{O}$ $\otimes$ $\mathbb{C}$	True
71	the norms correspond: N_oct(ϕ(x)) == N_split(x) (symbolic)	True

4.1 The exact mirror of T16: the Euclidean Clifford algebra Cl(8,0)

#	check	result
72	Cl(8,0): (T16O[A].T16O[B] + T16O[B].T16O[A])/2 == gO[[A+1,B+1]] ID16	True
73	S44 == diag(σ, σ)	True
74	T16O from the 4+4 matrices: T16O[A] == T16[A] (A = 0..3), T16O[k] == T16[k].S44 (k = 4..7)	True
75	T16O[8] == T16[8] == diag(-ID8, ID8): same chirality matrix and same PL44, PR44 as 4+4.nb	True

4.2 The REAL Clifford algebra of 7+1 spacetime: Γ17, Cl(1,7) $\cong$ $\mathbb{R}$(16)

#	check	result
76	7+1 REAL Clifford relation: (Γ17[A].Γ17[B] + Γ17[B].Γ17[A])/2 == η17[[A+1,B+1]] ID16 == -η71[[A+1,B+1]] ID16	True
77	Γ17[A] are REAL: every entry is 0 or ±1	True
78	block form: Γ17[0] == {{0,ID8},{ID8,0}}, Γ17[i] == {{τO[i],0},{0,τObar[i]}}	True
79	built from the real 4+4 matrices: Γ17[0] == T16[0], Γ17[i] == T16[0].T16[i] (i = 1,2,3), Γ17[k] == T16[0].T16[k].S44 (k = 4..7)	True
80	Γ17[0] symmetric, Γ17[i] antisymmetric, so Transpose[Γ17[A]] == Γ17[0].Γ17[A].Γ17[0] (real Dirac adjoint)	True
81	the 256 products Γ17[A1]...Γ17[Ak] (A1 < ... < Ak) are linearly independent: Cl(1,7) $\cong$ $\mathbb{R}$(16)	True
82	grades: binomial(8,k) == {1,8,28,56,70,56,28,8,1}, total 256 (binomial(8,4) == 70, not 72)	True
83	Tr[product] == 0 for every non-empty product, so Tr[M]/16 is the scalar part of M	True
84	irreducible over $\mathbb{R}$: only multiples of ID16 commute with all Γ17[A] (Schur; real commutant of dimension 1)	True

#	check	result
85	universal property: Γ17(v).Γ17(v) == (v.η17.v) ID16 for a symbolic vector v = X	True

4.3 The complexified 7+1 realization in the exact block form of 4+4.nb: Γ71

#	check	result
86	7+1 COMPLEX Clifford relation, 4+4.nb sign: (Γ71[A].Γ71[B] + Γ71[B].Γ71[A])/2 == η71[[A+1,B+1]] ID16	True
87	Γ71[0] == I T16[0] and Γ71[k] == T16O[k] (k = 1..7): only the time component is complex	True
88	S = (ID16 - I Γ17[0])/Sqrt[2] is unitary and Γ71[A] == S.(I Γ17[A]).Inverse[S]	True

4.4 Theorem: with η71 there is no real 16×16 realization, because Cl(7,1) ≅ ℍ(8)

#	check	result
89	Schur: the intertwiners M with M.Conjugate[Γ71[A]] == Γ71[A].M form a 1-dimensional space	True
90	... spanned by the volume element Γ17[8] == {{0,-ID8},{ID8,0}}, and Γ17[8].Γ17[8] == -ID16	True
91	indicator(Γ71) == -1: Cl(7,1) ≅ ℍ(8), NO real 16×16 realization of the η71 relation	True
92	indicator(Γ17) == +1: Cl(1,7) ≅ ℝ(16), real realization exists (Γ17)	True

4.5 A real realization of Cl(7,1) needs 32×32 matrices

#	check	result
93	REAL 32×32 realization of the η71 relation: (Γ71R32[A].Γ71R32[B] + Γ71R32[B].Γ71R32[A])/2 == η71[[A+1,B+1]] ID32	True
94	Qi, Qj, Qk commute with every Γ71R32[A]; Qi.Qi == Qj.Qj == Qk.Qk == -ID32; Qi.Qj == Qk == -Qj.Qi	True
95	the commutant of {Γ71R32[A]} has dimension exactly 4 and is spanned by ID32, Qi, Qj, Qk: it is ℍ	True

4.6 Volume elements and chirality

#	check	result
96	Γ17[8].Γ17[8] == -ID16 and Γ17[8] anticommutes with every Γ17[A]: no real chirality in 7+1	True
97	Γ71[8].Γ71[8] == -ID16 and χ71 = -I Γ71[8] == T16O[8] == T16[8] (real, diagonal)	True
98	7+1 chiral projectors in the Γ71 basis are exactly PL44, PR44 of 4+4.nb; χ71 anticommutes with every Γ71[A]	True
99	complex chiral projectors for Γ17: PL17 + PR17 == ID16, PL17.PL17 == PL17, PL17.PR17 == 0, rank 8 each	True

4.7 What the octonions do: Γ(x)² = Re(x x), and alternativity

#	check	result
100	Re(x x) == X.η17.X: the octonion square produces the 7+1 interval (symbolic)	True
101	octonion basis: Γ17e[0] == {{0,ID8},{ID8,0}} and Γ17e[i] == {{LO[i],0},{0,-LO[i]}}	True
102	alternativity in action: xImag(xImag ψ) == -|xImag|^2 ψ for the imaginary part xImag of x (symbolic)	True

#	check	result
103	T16O(x)^2 == N(x) ID16 (octonion norm) and T16(x)^2 == (X.η4488.X) ID16 (4+4.nb)	True

5.1 The Lorentz algebra so(1,7)

#	check	result
104	28 real generators Σ17[A,B] (A < B), linearly independent	True
105	[Σ17[A,B], Γ17[K]] == η17[[B+1,K+1]] Γ17[A] - η17[[A+1,K+1]] Γ17[B]	True
106	so(1,7): [Σ_AB, Σ_KL] == η_BK Σ_AL - η_AK Σ_BL - η_BL Σ_AK + η_AL Σ_BK	True
107	the complex structure J = Γ17[8] commutes with all Σ17[A,B] (Spin(1,7) acts complex-linearly on ℝ^16 = ℂ^8)	True

5.2 A boost along x1 and a rotation in the x4–x5 plane

#	check	result
108	S(β) and R(θ) are the exponentials: dS/dβ == Σ17[0,1].S, S(0) == ID16, dR/dθ == Σ17[4,5].R, R(0) == ID16, and S.Sinv == R.Rinv == ID16	True
109	S(β).Γ17[K].S(β)^-1 == Σ_B Λboost[[B+1,K+1]] Γ17[B]; Λboost is the boost along x1	True
110	R(θ).Γ17[K].R(θ)^-1 == Σ_B Λrot[[B+1,K+1]] Γ17[B]; Λrot is the rotation in the x4-x5 plane	True
111	both Λ preserve the 7+1 metric: Transpose[Λ].η17.Λ == η17	True

5.3 The 7+1 Dirac operator squares to the 7+1 wave operator

#	check	result
112	(Γ17^A ∂_A)^2 Ψ == (∂0^2 - ∂1^2 - ... - ∂7^2) Ψ for 16 arbitrary functions of x0..x7	True

5.4 Spinors on the mass shell: real amplitudes

#	check	result
113	p is on the mass shell (p.η17.p == m^2) and pslash.pslash == m^2 ID16	True
114	(pslash - m) w == 0 has 8 linearly independent REAL solutions w	True
115	Ψ = w Exp[-I p.x] solves (I Γ17^A ∂_A - m) Ψ == 0	True

5.5 Real spinor fields: massless with Γ17, massive with the real 32×32 matrices

#	check	result
116	the matrices anticommuting with every Γ17[A] are exactly the multiples of Γ17[8], and (c Γ17[8])^2 == -c^2 ID16: no real J with J.J == +ID16	True
117	null momentum k = (1,1,0,0,0,0,0,0): k.η17.k == 0, kslash.kslash == 0, and kslash.w == 0 has 8 REAL solutions	True
118	Ψ = w Cos[k.x] is a REAL solution of the real massless Dirac equation Γ17^A ∂_A Ψ == 0	True
119	(Γ71R32^A ∂_A)^2 Ψ == -(∂0^2 - ∂1^2 - ... - ∂7^2) Ψ for 32 arbitrary functions of x0..x7	True
120	p.η71.p == -m^2 and pslash32.pslash32 == -m^2 ID32	True
121	Ψ = w Cos[p.x] + (pslash32.w/m) Sin[p.x], with 32 arbitrary REAL constants w, solves the REAL massive equation (Γ71R32^A ∂_A - m) Ψ == 0	True
122	... and Ψ obeys the Klein-Gordon equation (∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) Ψ == 0	True

5.6 All nine signatures p + q = 8, and the corrected table of [1B]

#	check	result
123	split analogue: Re(x x) of split octonions == X.η53.X, and Γ53 realizes Cl(5,3) with REAL 16×16 matrices	True
124	Γ08[A] = T16O[8].T16O[A] realizes Cl(0,8) with REAL 16×16 matrices	True
125	gens[p] satisfy the Clifford relation of signature (p, 8-p) for p = 0..8	True
126	classification: ℝ(16) for (8,0),(5,3),(4,4),(1,7),(0,8); ℍ(8) for (7,1),(6,2),(3,5),(2,6)	True
127	every ℝ(16) case has explicit REAL 16×16 generators in this notebook (T16O, Γ53, T16, Γ17, Γ08)	True
128	ℂ(16) cannot occur: its real dimension 2*16^2 = 512 differs from 2^8 = 256	True

7.1 The eight real Dirac matrices and the Dirac α, β

#	check	result
129	Dirac matrices: Γ17up[0] == Γ17[0], Γ17up[i] == -Γ17[i]; all REAL, every entry 0 or ±1	True
130	Dirac adjoint: ConjugateTranspose[Γ17up[A]] == Γ17[0].Γ17up[A].Γ17[0] for A = 0..7	True
131	βD == {{0,ID8},{ID8,0}} and αD[i] == {{0,τO[i]},{-τO[i],0}}: real and symmetric	True
132	αD[i].αD[j] + αD[j].αD[i] == 2 δij ID16, αD[i].βD + βD.αD[i] == 0, βD.βD == ID16	True
133	ω17 = Γ17[8] == {{0,-ID8},{ID8,0}} is real, ω17.ω17 == -ID16, and ω17 anticommutes with every Γ17[A]	True

7.2 The field dirac16complex and its Dirac equation

#	check	result
134	(I Γ17up[A] ∂_A + m)(I Γ17up[A] ∂_A - m) dirac16complex == -(∂0^2 - ∂1^2 - ... - ∂7^2 + m^2) dirac16complex (16 arbitrary functions)	True

7.3 Hamiltonian form

#	check	result
135	Γ17[0].(I Γ17up[A] ∂_A - m) ψ == I ∂0 ψ - h ψ with h = -I αD[i] ∂_i + βD m (16 arbitrary functions)	True

7.4 Plane waves and the real spinors u_s, v_s

#	check	result
136	h(p) == {{0, M}, {Transpose[M], 0}} with the REAL 8×8 matrix M = m ID8 + p_i τO[i]	True
137	M.Transpose[M] == Transpose[M].M == E^2 ID8 and h(p).h(p) == E^2 ID16, E^2 = m^2 + p^2	True
138	h(p).u_s == E u_s and h(-p).v_s == -E v_s for s = 1..8	True
139	(pslash - m).u_s == 0 and (pslash + m).v_s == 0, pslash = E Γ17[0] + p_i Γ17[i]	True
140	u_s.u_r == v_s.v_r == 2E δsr, and u_s(p).v_r(-p) == 0	True
141	plane waves: u_s Exp[-I(E x0 - p.x)] and v_s Exp[+I(E x0 - p.x)] solve (I Γ17up[A] ∂_A - m) ψ == 0	True
142	example p = (2,2,4,0,0,0,0), m = 1: E = 5 and the eight u_s are real (entries in Sqrt[5] ℚ) and solve h(p).u_s == 5 u_s	True

7.5 Spin sums, completeness, and the diagonalization of h

#	check	result
143	spin sums: Σ_s u_s ubar_s == pslash + m and Σ_s v_s vbar_s == pslash - m	True
144	completeness: Σ_s u_s(p) u_s(p)^T + Σ_s v_s(-p) v_s(-p)^T == 2E ID16	True
145	U = (u_1..u_8, v_1(-p)..v_8(-p))/Sqrt[2E] is orthogonal and Transpose[U].h(p).U == diag(E ID8, -E ID8)	True

7.6 Grassmann algebra: anticommuting components, Lagrangian, conserved current

#	check	result
146	Grassmann: θ1 θ2 == -θ2 θ1, θ1 θ1 == 0, associativity, and ∂/∂θ2 (θ1 θ2 θ3) == -θ1 θ3	True
147	the three plane waves U1, U2, U3 solve the Dirac equation with m = 1	True
148	Grassmann field: ψ_a ψ_b + ψ_b ψ_a == 0 for all a, b (in particular ψ_a^2 == 0)	True
149	current conservation ∂_A (ψbar Γ17up[A] ψ) == 0 for the Grassmann-valued solution ψ = θ1 U1 + θ2 U2 + θ3 U3	True

7.7 Quantization: mode expansion and canonical anticommutators

#	check	result
150	canonical anticommutator from the mode algebra: {ψ_a(x0,x), ψ_b(x0,y)^†} == δab (1/V) Σ_p Exp[I p.(x - y)] (five momenta, symbolic x, y)	True

7.8 Charge conjugation

#	check	result
151	ψ -> ψc = ω17.ψ^*: (I Γ17up[A] ∂_A - m)(ω17.ψ^*) == ω17.((I Γ17up[A] ∂_A - m) ψ)^* for arbitrary ψ = ψre + I ψim	True
152	(pslash + m).ω17.u_s == 0 and ω17.u_s == Σ_r v_r (-Transpose[M]/E)_rs; ω17.ω17 == -ID16 (no Majorana spinors)	True

7.9 Lorentz covariance

#	check	result
153	boost at β = 2 log 2: S == 5/4 ID16 + 3/4 Γ17[0].Γ17[1], p' = Λ.p == (55/8, -41/8, 2, 4, 0, 0, 0, 0) is on shell, (pslash' - m).S.u_s(p) == 0	True

7.10 Chirality: no real chirality matrix, complex chiral projections

#	check	result
154	χ17 = I ω17: Hermitian, χ17.χ17 == ID16, anticommutes with every Γ17[A], commutes with every Σ17[A,B]	True
155	PL17, PR17 are complex (not real); Γ17[A].PL17 == PR17.Γ17[A]; ConjugateTranspose[PL17].Γ17[0] == Γ17[0].PR17	True
156	EL, ER: χ17.EL == -EL, χ17.ER == ER, EL^†.EL == ER^†.ER == ID8, EL^†.ER == 0, EL.EL^† == PL17, ER.ER^† == PR17, W unitary	True

#	check	result
157	φL == (ψup + I ψdown)/Sqrt[2], φR == (I ψup + ψdown)/Sqrt[2], PL17.ψ == EL.φL, PR17.ψ == ER.φR, ψ == EL.φL + ER.φR	True

7.11 The two coupled 8-component equations

#	check	result
158	σW[i] = I τO[i]: Hermitian, σW[i].σW[j] + σW[j].σW[i] == 2 δij ID8	True
159	chiral (Weyl) form: W^†.h(p).W == {{-σW.p, m ID8}, {m ID8, σW.p}} (symbolic p)	True
160	EL^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 + τO[i] ∂_i) φL - m φR (16 arbitrary functions)	True
161	ER^†.Γ17[0].(I Γ17up[A] ∂_A - m) ψ == (I ∂0 - τO[i] ∂_i) φR - m φL (16 arbitrary functions)	True
162	Klein-Gordon for each 8-component field: (I ∂0 - τO[i] ∂_i)(I ∂0 + τO[j] ∂_j) f == -(∂0^2 - ∂1^2 - ... - ∂7^2) f (8 arbitrary functions)	True
163	plane-wave example p = (2,2,4,0,0,0,0), m = 1, E = 5: (E + σW.p) φL == m φR and (E - σW.p) φR == m φL for all eight u_s	True

8.1 The self-coupled field: Lagrangian, densities, Hamiltonian

#	check	result
164	the densities ρS, ρP, j^A, j5^A are real: Γ17[0].op is Hermitian for op = ID16, I χ17, Γ17up[A], Γ17up[A].χ17	True
165	18 vertex pairs; lower vertex == η17[[A+1,A+1]] upper vertex in the V and A channels	True

8.2 The nonlinear field equation (Grassmann left derivatives)

#	check	result
166	nonlinear field equation: left derivative ∂L_int/∂ψbar_a == -[gS (ψbar ψ) ψ + gP (ψbar Iχ17 ψ) Iχ17 ψ + gV (ψbar Γ17[A] ψ) Γ17up[A] ψ + gA (ψbar Γ17[A]χ17 ψ) Γ17up[A]χ17 ψ]_a (32 Grassmann generators)	True

8.4 The 16-component Kohn–Sham equations

#	check	result
167	h_KS(p) is Hermitian for real S, P, V_A, A_A (symbolic momentum and potentials)	True
168	plane wave: Γ17[0].(pslash(ε,p) - m - S - I χ17 P - Γ17up[A](V_A + χ17 A_A)) == ε ID16 - h_KS(p)	True
169	Hartree potentials: ∂E_H/∂φ_a^* == (v_H.φ)_a, v_H = h_KS - h with S = gS ρS, P = gP ρP, V_A = gV j_A, A_A = gA j5_A (32 variables)	True

8.5 Exchange: Wick's theorem for contact couplings

#	check	result
170	Wick's theorem in the 560-dimensional Fock space: <Φ| :(ψbar o1 ψ)(ψbar o2 ψ): |Φ> == tr(o1 R) tr(o2 R) - tr(o1 R o2 R) for all 18 vertex pairs	True

8.6 The exchange self-energy and the Fierz identities

#	check	result
171	exchange self-energy: d/dt E_x(R + t dR) at t = 0 == tr(Σ_x.dR) and E_x == tr(Σ_x.R)/2 (random complex R, dR; symbolic couplings)	True
172	the 256 products Γ_K are orthogonal matrices with tr(Inverse[Γ_K].Γ_L) == 16 δKL	True
173	Fierz completeness: Σ_K (Γ_K)_ab (Inverse[Γ_K])_cd == 16 δad δcb (as the 256×256 matrix identity Transpose[F].F == 16 ID)	True
174	Fierz rearrangement of every exchange term: tr(o1 R o2 R) == (1/16) Σ_K tr(Inverse[Γ_K].R) tr(o1.Γ_K.o2.R) for all 18 vertex pairs (random complex R)	True

8.7 Example: uniform matter (no sea), Hartree and Hartree–Fock

#	check	result
175	uniform matter: tr R == ρS, tr(Γ17up[0].R) == ρ, and the densities ρP, j^i, j5^A vanish	True
176	the occupied plane waves give this R: (1/2) Σ_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) == (E Γ17[0] + m ID16)/(2E)	True
177	uniform exchange energy: E_x(R) == -gS/32 (ρS^2 + ρ^2) + gP/32 (ρS^2 - ρ^2) - gV/32 (8ρS^2 - 6ρ^2) + gA/32 (8ρS^2 + 6ρ^2)	True
178	uniform exchange self-energy: Σ_x == S_x ID16 + V_x Γ17[0] with S_x == ∂E_x/∂ρS and V_x == ∂E_x/∂ρ	True
179	uniform Hartree-Fock == Hartree with the Fierz-transformed couplings: gS/2 ρS^2 + gV/2 ρ^2 + E_x == G_S/2 ρS^2 + G_V/2 ρ^2 (gP, gA arbitrary)	True
180	c7 == 8 |S^6|/(2 Pi)^7 == 1/(15 Pi^4), |S^6| == 2 Pi^(7/2)/Gamma[7/2] == 16 Pi^3/15, and ρ == c7 kF^7/7 == kF^7/(105 Pi^4)	True
181	closed forms: d/dkF of the integrals gives the integrands, both vanish at kF = 0, and dε_kin/dm* == ρS	True
182	Hartree gap equation m* == m + gS ρS(m*) solved to 45 digits; m* is a stationary point of ε(m*) at fixed kF	True
183	total energy == sum of KS eigenvalues - E_H (double counting) == ε_kin - (m* - m) ρS + E_H	True
184	Hugenholtz-Van Hove at saturation: E/A == ε_F == dε/dρ; E/A - m < 0 and E/A at 0.95 kF0 and 1.05 kF0 is higher	True
185	Hugenholtz-Van Hove away from saturation: μ == dε/dρ == ε_F at kF = 1	True
186	Hartree-Fock at kF = 3/4 (gP = gA = 0): G_S == -98750, G_V == 20000; m* - m == gS ρS + S_x, V_0 == gV ρ + V_x; μ == ε_F	True

9.1 Chiral densities

#	check	result
187	chiral densities: ρS == κ + κ^*, ρP == I(κ - κ^*), j^A == jL^A + jR^A, j5^A == jR^A - jL^A (symbolic chiral orbitals)	True
188	the chiral blocks behind these densities: EL^†.βD.ER == ID8, EL^†.βD.EL == 0, EL^†.αD[i].EL == -σW[i], ER^†.αD[i].ER == σW[i], EL^†.αD[i].ER == 0	True

9.2 The Hartree energy in chiral variables

#	check	result
189	E_H (16-component form, chiral orbitals) == gS/2 (κ + κ^*)^2 - gP/2 (κ - κ^*)^2 + (gV + gA)/2 (jL.jL + jR.jR) + (gV - gA) jL.jR	True

9.3 The chiral Kohn–Sham equations

#	check	result
190	chiral KS Hamiltonian: W^†.h_KS(p).W == {{V^L_0 - σW.(p + V^L), M}, {M^*, V^R_0 + σW.(p + V^R)}}, M = m + S + I P, V^L = V - A, V^R = V + A	True
191	Hartree (Wirtinger): M - m == ∂e_H/∂κ == gS ρS + I gP ρP, V^L_A == ∂e_H/∂jL^A; V^L_A == V_A - A_A and V^R_A == V_A + A_A with V_A = gV j_A, A_A = gA j5_A	True
192	chiral KS potentials are derivatives: ∂E_H/∂φL^* == (V^L_0 - σW[i] V^L_i) φL + (M - m) φR and ∂E_H/∂φR^* == (V^R_0 + σW[i] V^R_i) φR + (M - m)^* φL	True

9.4 Chiral symmetry

#	check	result
193	e^(Iθχ17) == cosθ ID16 - sinθ ω17 is REAL and orthogonal; U.EL == e^(-Iθ) EL, U.ER == e^(Iθ) ER	True
194	U^†.βD.U == βD.(cos 2θ ID16 + I sin 2θ χ17); vector and axial densities invariant	True
195	E_H(e^(Iθχ17) ψ) - E_H(ψ) == (gS - gP)/2 [(e^(4Iθ) - 1) κ^2 + (e^(-4Iθ) - 1) κ^*2]: U(1)_A symmetric iff gP == gS	True

9.5 Uniform matter in chiral form

#	check	result
196	uniform matter in chiral form: κ == ρS/2, jL^0 == jR^0 == ρ/2, jL^i == jR^i == 0	True
197	(h_χ - V_0)^2 == (p^2 + m*^2) ID16: eigenvalues V_0 ± Sqrt[p^2 + m*^2], eight times each	True

9.6 The chiral vacuum with the Dirac sea: spontaneous chiral symmetry breaking

#	check	result
198	|S^6|/(2 Pi)^7 == 1/(120 Pi^4); J(0) == Λ^6/(720 Pi^4); g_c == -1/(8 J(0)) == -90 Pi^4/Λ^6; J(M) -> J(0) and K(M) -> K(0) as M -> 0	True
199	m = 0, g = 6/5 g_c: the gap equation 1 + 8 g J(|M|) == 0 has the root |M| (32 digits); M == 2 g κ^* with κ == -4 M J; E is stationary there and lower than E(0)	True
200	degenerate chiral vacua for m = 0: E(|M| e^(I ϕ)) == E(|M|) for ϕ = 1, 2, 4 (the phase is the Goldstone direction)	True
201	explicit breaking m = 1/100: the real M == m - 8 g J(M) M solves the gap equation, E is stationary there, and every complex phase raises E	True
202	the sea at one momentum: h_χ^2 == E^2 ID16; Π_- is a projector of rank 8; κ_p == tr(Π_- RL block) == -4 M^*/E; tr(Π_-.h0) == -8E + 8 Re[(M - m) M^*]/E	True

11.2 The cosmological metric of 7+1 spacetime

#	check	result
203	cosmological metric: sectional curvatures K_0i = -(Ha'+Ha^2), K_0k = -(Hb'+Hb^2), K_ij = -Ha^2, K_kl = -Hb^2, K_ik = -Ha Hb	True
204	cosmological metric: R^cd_ab is non-zero only for {c,d} = {a,b}	True
205	Einstein tensor: G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2, G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb, G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb	True
206	scalar curvature R = -2 (3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 Ha Hb)	True

11.3 Symmetries of the Riemann tensor and the contracted Bianchi identity

#	check	result
207	inhomogeneous metric, exactly at (x0, x4) = (1/2, 1/3): R_rsmn = -R_srmn = -R_rsnm = R_mnrs and R^r_[smn] = 0 (first Bianchi identity)	True
208	inhomogeneous metric: contracted Bianchi identity nabla_mu G^mu_nu == 0, exactly at (x0, x4) = (1/2, 1/3)	True
209	cosmological metric: contracted Bianchi identity nabla_mu G^mu_nu = 0 (arbitrary a, b)	True

12.1 The generalized Kronecker delta

#	check	result
210	generalized Kronecker delta: contraction identity delta^{m1..mp r}_{n1..np r} = (8-p) delta^{m1..mp}_{n1..np} (p = 1..4, 60 random index sets each)	True
211	generalized Kronecker delta with 9 upper indices vanishes in 8 dimensions (100 random index sets)	True

12.3 The Lovelock tensors with abstract indices: the generalized Kronecker delta kδ

#	check	result
212	xAct: Lovelock1 == R_a^b - (1/2) delta_a^b R, the Einstein tensor (abstract indices)	True
213	xAct: trace of Lovelock2 == -2 L_2 with L_2 = R^2 - 4 R_cd R^cd + R_cdfi R^cdfi (abstract indices)	True

12.4 Components with xAct: CD = LC[g]

#	check	result
214	xAct, cosmological metric: Lovelock1 == EinsteinCD == the Einstein tensor of the curvature engine of section 11.1	True

12.5 An independent computation: the pair-set algorithm

#	check	result
215	pair-set algorithm == brute-force generalized-delta contraction for E^(1) (cosmological metric)	True
216	pair-set algorithm == xAct (section 12.4) for E^(1), E^(2), E^(3) of the cosmological metric (two independent computations)	True

12.6 The Lovelock tensors of the cosmological metric

#	check	result
217	E^(1) == Einstein tensor G and L_1 == R (cosmological metric)	True
218	trace identity E^(k)mu_mu == -((8-2k)/2) L_k for k = 0..4	True
219	E^(4) == 0 identically in 8 dimensions, while the Euler density L_4 != 0	True
220	every E^(k) is diagonal, E^(k)1_1 = E^(k)2_2 = E^(k)3_3 and E^(k)4_4 = ... = E^(k)7_7	True

#	check	result
221	E^(2) == Lanczos tensor and L_2 == R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric)	True
222	E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2 Hb + 12 Ha Hb^2 + Hb^3) and E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb)	True
223	divergence-free: d0 E^0_0 + 3Ha (E^0_0 - E^1_1) + 4Hb (E^0_0 - E^4_4) == 0 for k = 1, 2, 3 (arbitrary a, b)	True
224	variation of nL a^3 b^4 L_k: EL_nL = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4 E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4 (k = 1..4)	True

12.7 An inhomogeneous metric: the Lovelock tensors are divergence-free (exact check at a point)

#	check	result
225	inhomogeneous metric: nabla_mu E^(k)mu_nu == 0 EXACTLY at (x0, x4) = (1/2, 1/3) for k = 1, 2, 3	True
226	inhomogeneous metric: E^(1) == G at the point, and E^(k)_mu nu symmetric (k = 1, 2, 3)	True
227	inhomogeneous metric: E^(2) == Lanczos tensor at the point (exact)	True

12.8 A plane gravitational wave: only the Einstein tensor survives

#	check	result
228	pp-wave: E^(1)_mu nu == +(1/2)(d2^2 + d3^2) fPP l_mu l_nu	True
229	pp-wave: E^(2) == E^(3) == 0 for every profile fPP	True
230	pp-wave: xAct (section 12.4) == pair-set algorithm for E^(1), E^(2), E^(3) (a metric that is not diagonal; arbitrary profile fPP)	True

12.9 Exponential vacuum solutions: eternal inflation of ordinary space and deflation of the auxiliary space

#	check	result
231	exponential expansion a = e^{H x0}, b = e^{qH H x0}: xAct Lovelock_k == the closed forms of section 12.6 with Ha = H, Hb = qH H, Ha' = Hb' = 0 (k = 1, 2, 3)	True
232	exponential expansion: three distinct vacuum equations, related by 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) == 0	True
233	exponential vacuum solutions: (w2, w3) solve all 64 components for every qH (generic qH) and every Lambda	True
234	eternal inflation (a = e^{H x0}) with deflation (b = e^{-H x0}): w2 == Lambda/4 - w1/2, w3 == w1/24 - Lambda/36; for Lambda = 0: w2/w1 == -1/2, w3/w1 == 1/24	True

13.2 The cosmological metric: frame, spin connection, Dirac equation

#	check	result
235	cosmological metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True
236	cosmological metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True
237	cosmological metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True
238	cosmological metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True
239	cosmological metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True

#	check	result
240	cosmological metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True
241	cosmological metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True
242	cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True
243	cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True
244	cosmological metric: Omega_0 == 0, Omega_i == (a'/2) Gamma17[i].Gamma17[0], Omega_k == (b'/2) Gamma17[k].Gamma17[0]	True
245	cosmological metric: Gamma^mu Omega_mu == (3a'/(2a) + 2b'/b) Gamma17[0]	True
246	current identity: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi] (psi, psibar arbitrary)	True
247	Lichnerowicz formula: (Gamma^mu D_mu)^2 psi == (g^mu nu nabla_mu D_nu - R/4) psi	True
248	rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi == (a^3 b^4)^(-1/2) Gamma^mu d_mu chi	True
249	mode equation: Gamma17[0].(i Gamma^mu D_mu - m) psi == (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} (i dc/dx0 - h(x0) c) for psi = (a^3 b^4)^(-1/2) e^{i(p.x+q.y)} c(x0)	True

13.3 The pp-wave and the inhomogeneous metric: frames that are not diagonal and not homogeneous

#	check	result
250	pp-wave: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True
251	pp-wave: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True
252	pp-wave: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True
253	pp-wave: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True
254	pp-wave: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True
255	pp-wave: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True
256	pp-wave: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True
257	pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True
258	pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True
259	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True
260	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True
261	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True
262	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True
263	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True
264	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True
265	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True
266	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True
267	inhomogeneous metric (exact at (x0, x4) = (1/2, 1/3)): (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True

13.4 Conformal weight 7/2 of the massless field

#	check	result
268	conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) == C^(-9/2) Gamma17up[A] d_A f for g = C(x0)^2 eta17	True

13.5 Cartan's first structure equation, a static metric with a radial coordinate, and the coupling to a gauge field

#	check	result
269	Cartan's first structure equation de^A + omega^A_B ^ e^B == 0 (no torsion): cosmological, pp-wave and inhomogeneous frames	True
270	structure coefficients: omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA) with C^A_BC = e_B^mu e_C^nu (d_mu e^A_nu - d_nu e^A_mu): cosmological and inhomogeneous frames	True
271	static radial metric: eta17 e e == g and {Gamma_mu, Gamma_nu} == 2 g_mu nu ID16 (generalized frame matrices, real)	True
272	static radial metric: canonical spin connection antisymmetric, omega_mu AB == -omega_mu BA	True
273	static radial metric: vielbein postulate d_mu e^A_nu - {l mu nu} e^A_l + omega_mu^A_B e^B_nu == 0	True
274	static radial metric: TOTAL covariant derivative of the frame matrices vanishes, d_mu Gamma_nu - {l mu nu} Gamma_l + [Omega_mu, Gamma_nu] == 0	True
275	static radial metric: constant frame matrices are covariantly constant, omega_mu^A_B Gamma17up[B] + [Omega_mu, Gamma17up[A]] == 0	True
276	static radial metric: Omega_mu real and Gamma17[0].Transpose[Omega_mu].Gamma17[0] == -Omega_mu (so D_mu psibar = d_mu psibar - psibar Omega_mu)	True
277	static radial metric: spin curvature F_mu nu == (1/4) R_AB mu nu Gamma17up[A].Gamma17up[B]	True
278	static radial metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic == 0	True
279	static radial metric: (1/2) Gamma^mu Gamma^nu F_mu nu == -(R/4) ID16	True
280	static radial metric: Cartan's first structure equation and omega_C,AB == -(1/2)(C_ACB + C_BAC - C_CBA)	True
281	static radial metric: Omega_0 == (f'/(4 Sqrt[f] Sqrt[h])) Gamma17[1].Gamma17[0], Omega_1 == 0, Omega_k == (1/(2 Sqrt[h])) Gamma17[1].Gamma17[k] (k = 2..7)	True
282	static radial metric: Gamma^mu Omega_mu == (1/Sqrt[h]) (f'/(4 f) + 3/r) Gamma17up[1]	True
283	static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection, Gamma^mu D_mu psi == f^(-1/4) r^(-3) Gamma^mu d_mu chi	True
284	U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) == e^{i q alpha} Gamma^mu (D_mu - i q A_mu) psi with A'_mu = A_mu + d_mu alpha	True
285	current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) == sqrt|g| [(D_mu psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu	True

14.1 Primordial epoch near x0 = 0: Kasner solutions

#	check	result
286	Kasner metric: the Einstein tensor computed directly from the metric equals the closed form (x0^2 G^mu_mu, mu = 0, 1, 4)	True
287	Einstein Kasner: (p, q) = ((1+2Sqrt[2])/7, (2-3Sqrt[2])/14) and ((1-2Sqrt[2])/7, (2+3Sqrt[2])/14) solve E^(1) == 0	True
288	Gauss-Bonnet Kasner: 3 real roots of the cubic; (p, (3-3p)/4) solve E^(2) == 0	True
289	cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) == 0; E^(3) == 0 identically for q = 0 and for p = 0	True

14.2 The mode equation in two dimensions

#	check	result
290	2x2 reduction: A, B, betaD square to ID16 and anticommute; Zc = A.B.betaD: Zc.Zc == -ID16, Transpose[Zc] == -Zc, Zc commutes with A, B, betaD	True

14.3 Pair creation in the primordial epoch (Einstein–Kasner branch p = 0.547, q = −0.160)

#	check	result
291	Kasner pair creation: |alpha|^2 + |beta|^2 == 1 (to 10^-8) for (|p|, |q|) = (1,1), (2,1), (1,3)	True
292	Kasner scaling: (|p|, |q|) = (1.4302.., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to 10^-6)	True
293	16x16 evolution == 4 copies of the 2x2 evolution with +m sigma3 and 4 with -m sigma3 (Kasner, m = 0.3, |p| = 1, |q| = 1.5, x0 from 0.05 to 40): average |beta|^2 agrees to 10^-8; unitarity; the two classes differ	True
294	isotropic expansion, massless: no pair creation (|beta|^2 < 10^-20)	True

14.4 Modern epoch: a static auxiliary space

#	check	result
295	static auxiliary space (Hb = 0): E^(1) = (3Ha^2, 2Ha' + 3Ha^2, 3Ha' + 6Ha^2), E^(2) = (0, 0, -12 Ha^2 (Ha' + Ha^2)), E^(3) = 0	True

14.5 Far future: de Sitter expansion and thermal pair creation

#	check	result
296	8D de Sitter: E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (all diagonal entries)	True
297	de Sitter over a static auxiliary space: E^(1) = (3, 3, 6) H^2, E^(2) = (0, 0, -12) H^4, E^(3) = 0	True
298	de Sitter pair creation: |beta|^2 == 1/(Exp[2 Pi M/H] + 1) to 10^-7, M/H = 0.25, 0.5, 1	True
299	integrator benchmark (Landau-Zener): |beta|^2 == Exp[-Pi kappa^2] to 10^-6	True

15.1 The nine channels

#	check	result
300	256 multi-indices in 9 ranks with 1, 8, 28, 56, 70, 56, 28, 8, 1 members	True
301	every bilinear psibar Theta^K psi is real: Gamma17[0].Theta^K Hermitian for all 256 K	True
302	Theta_K.Theta^K == ID16 and Tr[Theta_K.Theta^J] == 16 delta_KJ	True
303	Hodge duality: Theta^K.omega17 == (phase) Theta^(complement of K), |phase| = 1	True
304	the couplings of section 8 are four of the nine: I_0 == (psibar psi)^2, I_1 == (psibar Gamma^A psi)(psibar Gamma_A psi), I_8 == -(psibar i chi17 psi)^2, I_7 == -(axial invariant)	True
305	local Lorentz invariance: I_r(S psi, psibar S^-1) == I_r(psi, psibar) for r = 0..8 (random finite Lorentz transformation), and S^-1 == Gamma17[0].S^dagger.Gamma17[0]	True

15.2 The 9×9 Fierz matrix and Hartree–Fock with nine channels

#	check	result
306	Fierz matrix: closed form == (1/256) Sum_{K in r} Tr[Theta_K.Theta_J.Theta^K.Theta^J] for the first and last J of every rank s	True
307	F.F == IdentityMatrix[9]; eigenvalues: five +1 and four -1	True
308	Fierz identity for commuting spinors (random): Sum_{K in r} (a Theta_K b)(c Theta^K d) == Sum_s F_rs Sum_{J in s} (a Theta_J d)(c Theta^J b), r = 0..8	True
309	Hartree-Fock energy E_H + E_x == Sum_s (G_s/2) D_s with G = (1 - Transpose[F]) g (random density matrix and couplings)	True
310	uniform density matrix (rhoS ID16 + rho Gamma17[0])/16: D_0 == rhoS^2, D_1 == rho^2, D_2 .. D_8 == 0	True
311	section 8 recovered: with g = (gS, gV, 0, 0, 0, 0, 0, -gA, -gP) the exchange energy of uniform matter is that of section 8.7	True

#	check	result
312	the Hartree-Fock couplings of section 8.7: gS = -100000, gV = 10000 give G_0 = -98750, G_1 = 20000	True

15.3 The covariant nonlinear field equation

#	check	result
313	nonlinear field equation: dL_self/d psibar_a == -Sum_r g_r Sum_K (psibar Theta_K psi) (Theta^K psi)_a	True

15.4 Chirality in a gravitational field

#	check	result
314	chi17 commutes with every generator (Gamma17up[A].Gamma17up[B] - Gamma17up[B].Gamma17up[A])/4, hence with every Omega_mu	True
315	curved chiral equations: E_L^dag.Gamma17[0].(i Gamma^mu D_mu - m) psi == (i d0 + i(3Ha/2 + 2Hb) + tauO.d) phi_L - m phi_R, and the same for E_R with -tauO.d	True

16.1 Homogeneous Kohn–Sham matter in the cosmological spacetime

#	check	result
316	flat limit a = b = 1, kF = kFsat: m* == 0.3480973401... and E/A - m == -0.0283214279... (section 8.7), to 10^-11	True
317	flat limit: pa == pb == rho eps_F - eps (Gibbs-Duhem), which is 0 at the saturation density	True
318	energy-momentum conservation of the Kohn-Sham matter: eps' + 3Ha(eps + pa) + 4Hb(eps + pb) == 0 (a = 1 + 0.3x + 0.05x^2, b = 1/(1 + 0.2x), x0 = 0.7; relative 10^-8)	True

16.2 The three epochs

#	check	result
319	primordial: rho x0 == kFsat^7/(105 Pi^4) at every x0, and m* decreases monotonically to below 0.05 at x0 = 10^-4	True
320	modern (b = 1): eps a^3 tends to a constant and pa/eps -> 0 (pressureless in ordinary space)	True
321	far future (de Sitter, H = 0.1): rho == rho0 e^{-3 H x0} and 1 - m* decays like rho	True

16.3 Self-consistent evolution: Einstein–Lovelock gravity driven by the Kohn–Sham matter

#	check	result
322	the 11 and 44 Einstein-Lovelock equations are linear in (Ha', Hb'); the 00 equation contains neither (constraint)	True
323	self-consistent evolution: the constraint Sum alpha_k E^(k)0_0 - Lambda == kappa eps is preserved to 10^-7 (relative) for Einstein and for Einstein-Lovelock (alpha2 = 20, alpha3 = 500)	True

17.1 The fields, their quantum numbers, and anomaly cancellation

#	check	result
324	anomaly cancellation per generation (with nu_R): SU(3)^2U(1), SU(2)^2U(1), U(1)^3, grav^2U(1), SU(3)^3 all vanish; doublets per generation even	True
325	electric charges Q = T3 + Y: (u, d) = (2/3, -1/3), (nu, e) = (0, -1), u_R 2/3, d_R -1/3, e_R -1, nu_R 0	True
326	g* = 28 + (7/8) 90 = 106.75 for the Standard Model fields (without nu_R)	True

17.3 Electroweak symmetry breaking

#	check	result
327	gauge-boson mass matrix: eigenvalues g^2 v^2/4 (twice), (g^2 + g'^2) v^2/4 and 0 (photon)	True
328	tree level: m_W == g v/2, m_Z == Sqrt[g^2 + g'^2] v/2, rho == m_W^2/(m_Z^2 cos^2 theta_W) == 1, m_H^2 == 2 lambda v^2	True

17.4 Quark and lepton masses; CKM and PMNS mixing

#	check	result
329	CKM: unitary; Jarlskog invariant J = 3.12 x 10^-5 (PDG 2024: 3.12 +0.13 -0.12 x 10^-5); A, rhobar, etabar within the PDG uncertainties	True
330	Yukawa matrices: singular values x v/Sqrt[2] are the quark masses and |U_uL^dag.U_dL| == |V_CKM|	True
331	PMNS unitary; Dirac neutrino Yukawa Sqrt[2] U.diag(m)/v reproduces the masses (singular values)	True

17.5 The modern epoch on the brane

#	check	result
332	age of the universe from the brane Friedmann equation: 13.79 Gyr (Planck 2018: 13.787 +- 0.020 Gyr)	True
333	ADD relation: kappa8 rho4/V_perp == 8 Pi G4 rho4 with G4 = G8/V_perp (kappa8 = 8 Pi G8)	True

18. dirac16complex at the world brane: the 16 components as four brane Dirac spinors, and the portals

#	check	result
334	brane representation: R_A R_B + R_B R_A == 2 eta17_AB ID16 for R = (gamma_mu (x) 1_4, gamma5 (x) rho_k)	True
335	intertwiner: Gamma17[A].U == U.R_A for A = 0..7, and U is unitary	True
336	chi17 == U.(gamma5 (x) rho4567).U^dag; rho4567 Hermitian with eigenvalues {-1, -1, 1, 1}	True
337	bilinears at the brane: psibar psi' == Sum_a psibar_a psi'_a and psibar Gamma^mu psi' == Sum_a psibar_a gamma^mu psi'_a (mu = 0..3)	True

18.1 Kaluza–Klein masses on the auxiliary torus

#	check	result
338	KK operator: X.X == -|q|^2/b0^2 ID16 and X anticommutes with gamma_mu (x) 1_4	True
339	KK masses: Det[gamma.p (x) 1 + X - m] vanishes on p0^2 - |p|^2 == m^2 + |q|^2/b0^2 (40-digit arithmetic) and not off shell	True

18.2 The portals

#	check	result
340	portal mixing: 1 - N.N^dag == mu.mu^dag/m^2 to second order (non-unitary light mixing)	True

19. Neutrino flavour and flavour conversion

#	check	result
341	vacuum: P(a->b) doubly stochastic; equals delta - 4 Sum Re X sin^2 Delta + 2 Sum Im X sin 2Delta; CPT P(a->b) == P(bbar->abar)	True
342	V_CC(rho Y_e = 1 g/cm^3) == 7.63 x 10^-14 eV; matter evolution doubly stochastic; a flavour-universal potential has no effect	True
343	two-flavour MSW: sin^2 theta_m from diagonalization == formula, and sin^2 2theta_m == 1 at 2 E V == dm^2 cos 2theta	True
344	Sun: numerical P_ee == adiabatic MSW formula to 2 x 10^-3 at 0.3, 1, 5 and 10 MeV; P_ee(0.3 MeV) within 0.03 of the vacuum average; P_ee falls with energy	True
345	cosmic neutrino background: T_nu == 1.945 K and n == 112 cm^-3 per flavour	True
346	dirac16complex portal: non-unitary oscillations, Sum_b P(a->b) < 1 (zero-distance effect); decoupled limit == three-flavour vacuum result	True

4.3 Independent Python cross-checks

tools/verify_7p1_nb.py (parses the evaluated 7+1.nb; exit code 0)

```
input cells
    ...: 145
output cells
    ...: 382
message cells (must be 0)
    ...: 0
check outputs
    ...: 310
check outputs that are True
    ...: 310
check["..."] calls in the input cells
    ...: 310
$Version output
    ...: ['{"15.0.1 for Microsoft Windows (64-bit) (July 2, 2026)","Notebook$$41$124309`"}']
private notebook context
    ...: True
Length[checks]
    ...: 346
failedChecks == {}
    ...: True
allChecksPassed output True
    ...: True
tau[A] printed == 4+4 tau[A] (python)
```

```
    ...: True
tauO[A] printed == python tauO[A]
    ...: True
Gamma17[A] printed == python G17[A]
    ...: True
gridO printed == python octonion table
    ...: True
gridx44 printed == python split table (4+4.nb)
    ...: True
Lambda boost/rotation printed
    ...: True
8 real on-shell spinors printed solve (pslash - m) w = 0 (python check)
    ...: True
classification table printed
    ...: True
section 7: check outputs (all True)
    ...: 35 (True)
section 8: check outputs (all True)
    ...: 23 (True)
section 9: check outputs (all True)
    ...: 16 (True)
section 7: printed Sqrt[5] u_s(p) == python (5 e_s; M^T e_s) and (pslash - m) u_s == 0
    ...: True
section 8: printed closed forms of int k^6 m*/E and int k^6 E: d/dkF == integrands, zero at kF
    ...= 0 (sympy)                                              : True
sections 8-9: 23 printed numbers agree with tools/ref_dirac16.py to 17 digits
    ...: True
sections 8-9: G_S == -98750, G_V == 20000 printed
    ...: True
tools/ref_dirac16.py: all its boolean checks True
    ...: True
sections 0-6: the 166 input and text cells equal those of the backup
    ...: True
section 11: check outputs (all True)
    ...: 7 (True)
section 12: check outputs (all True)
    ...: 25 (True)
section 13: check outputs (all True)
    ...: 15 (True)
section 14: check outputs (all True)
    ...: 14 (True)
section 15: check outputs (all True)
    ...: 16 (True)
section 16: check outputs (all True)
    ...: 8 (True)
section 17: check outputs (all True)
    ...: 10 (True)
section 18: check outputs (all True)
    ...: 7 (True)
section 19: check outputs (all True)
    ...: 6 (True)
summary table: entries == Length[checks], all True
    ...: True
summary table: every printed check is listed
    ...: True
sections 0-10: the 368 input and text cells equal those of the backup taken before sections
    ...11-20                                                          : True
sections 14-19: 27 groups of printed numbers agree with the Python programs (relative 1e-6;
    ...rounded prints 1e-4; ODE and ill-conditioned results 1e-5) : True

VERDICT: ALL CROSS-CHECKS PASSED
```

tools/ref44.py (rebuilds 4+4.nb; exit code 0)

```
ζ7 = [-1, -1, -1, -1, 1, 1, 1, 1]
σ == ζ1ζ2ζ3 : True
{ζA,ζB}/2 == -η ID8 (A,B=1..7): True
-η_AA ζA == ζA^T : True
(τA τB+τB τA)/2 == η ID8 : True
(τA τB+τB τA)/2 == η ID8 : True
τ1..τ7 product == ID8: True
σ == τ4τ5τ6τ7: True
{T16A,T16B}/2 == η ID16 : True
T16[8] diag: [-1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1]
σ16 == diag(-σ,σ): True
unit: [sqrt(2)/2, 0, 0, 0, sqrt(2)/2, 0, 0, 0]  unit.σ.unit = 1
E^a_A[h] == (u σ u) Transpose[Table[η_BB τ[B-1].u]] for all h: True
eA[1] : ['eA[1]', 'eA[2]', 'eA[3]', 'eA[4]', 'eA[5]', 'eA[6]', 'eA[7]', 'eA[8]']
eA[2] : ['eA[2]', '-eA[1]', '-eA[4]', 'eA[3]', '-eA[6]', 'eA[5]', 'eA[8]', '-eA[7]']
eA[3] : ['eA[3]', 'eA[4]', '-eA[1]', '-eA[2]', '-eA[7]', '-eA[8]', 'eA[5]', 'eA[6]']
eA[4] : ['eA[4]', '-eA[3]', 'eA[2]', '-eA[1]', '-eA[8]', 'eA[7]', '-eA[6]', 'eA[5]']
eA[5] : ['eA[5]', 'eA[6]', 'eA[7]', 'eA[8]', 'eA[1]', 'eA[2]', 'eA[3]', 'eA[4]']
eA[6] : ['eA[6]', '-eA[5]', 'eA[8]', '-eA[7]', '-eA[2]', 'eA[1]', '-eA[4]', 'eA[3]']
eA[7] : ['eA[7]', '-eA[8]', '-eA[5]', 'eA[6]', '-eA[3]', 'eA[4]', 'eA[1]', '-eA[2]']
eA[8] : ['eA[8]', 'eA[7]', '-eA[6]', '-eA[5]', '-eA[4]', '-eA[3]', 'eA[2]', 'eA[1]']
```

tools/ref71.py (rebuilds 7+1.nb objects; exit code 0)

```
tauO antisymmetric (A>=1)                          : True
tauO[A]^2=-1 (A>=1)                                : True
tauO Euclid relation                               : True
tauObar tauO relation                              : True
tauO1..7 product                                   : [1, 1, 1, 1, 1, 1, 1, 1]
sigma commutes tau1..3, anticommutes tau4..7       : True
E orthogonal                                       : True
oct: unit two-sided                                : True
oct: e_i^2                                         : [-1, -1, -1, -1, -1, -1, -1]
oct: x xbar = N(x) 1                               : True
oct: N(xy)=N(x)N(y)                                : True
oct: conj(xy)=conj(y)conj(x)                       : True
oct: alternative (left,right,flexible)             : True
oct: Moufang identities                            : True
oct: #nonassociative basis triples                 : 168
oct: #noncommuting ordered basis pairs             : 42
e_A*e_B (O) = s_AB e_A*e_B (S)                     : True
T16O Euclid Cl(8,0)                                : True
T16O[A]==T16A44[A] (A<=3)                          : True
T16O[k]==T16A44[k].S44 (k>=4)                      : True
S44==diag(sigma,sigma)                             : True
T16O[8] diag                                       : [-1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1,
    ...1, 1, 1, 1, 1]
G17 Cl(1,7)                                        : True
G17 real                                           : True
G17[0] sym, G17[i] antisym                         : True
G17[8]^2                                           : -1
G17 blocks: G17[i]==diag(tauO,tauObar)             : True
G71 Cl(7,1) eta71                                  : True
G71[8]^2                                           : -1
G71 = Ssim (I G17) Ssim^-1                         : True
G71R32 Cl(7,1) real                                : True
```

tools/extra71.py (further identities; exit code 0)

```
Lambda boost (rows B = 0..7):
    ['cosh(beta)', '-sinh(beta)', '0', '0', '0', '0', '0', '0']
```

```
    ['-sinh(beta)', 'cosh(beta)', '0', '0', '0', '0', '0', '0']
    ['0', '0', '1', '0', '0', '0', '0', '0']
    ['0', '0', '0', '1', '0', '0', '0', '0']
    ['0', '0', '0', '0', '1', '0', '0', '0']
    ['0', '0', '0', '0', '0', '1', '0', '0']
    ['0', '0', '0', '0', '0', '0', '1', '0']
    ['0', '0', '0', '0', '0', '0', '0', '1']
Lambda rotation (rows B = 0..7):
    ['1', '0', '0', '0', '0', '0', '0', '0']
    ['0', '1', '0', '0', '0', '0', '0', '0']
    ['0', '0', '1', '0', '0', '0', '0', '0']
    ['0', '0', '0', '1', '0', '0', '0', '0']
    ['0', '0', '0', '0', 'cos(theta)', '-sin(theta)', '0', '0']
    ['0', '0', '0', '0', 'sin(theta)', 'cos(theta)', '0', '0']
    ['0', '0', '0', '0', '0', '0', '1', '0']
    ['0', '0', '0', '0', '0', '0', '0', '1']
indicator classification (signature, number of intertwiners, M.conj(M) for the normalized M,
    ...algebra):
    Cl(8,0): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(7,1): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(6,2): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(5,3): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(4,4): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(3,5): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(2,6): 1 intertwiner, M.conj(M) = -0.0625 * 1  ->  H(8)
    Cl(1,7): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
    Cl(0,8): 1 intertwiner, M.conj(M) = +0.0625 * 1  ->  R(16)
E0 == eta4488 . E44[8]                                                  : True
G53 Cl(5,3) real                                                        : True
G08 Cl(0,8) real                                                        : True
so(1,7): [S_AB,S_CD]=eta_BC S_AD - eta_AC S_BD - eta_BD S_AC + eta_AD S_BC: True
[S_AB,G_C]=eta_BC G_A - eta_AC G_B                                      : True
S Sinv = 1                                                              : True
S G_C S^-1 == sum_B L[B,C] G_B                                          : True
L^T eta L == eta                                                        : True
p on shell p.eta17.p == 1                                               : True
pslash^2 == 1                                                           : True
8 real solutions                                                        : True
classification                                                          : True
explicit quaternion commutant of G71R32                                 : True
G17e[0]==[[0,1],[1,0]]x1, G17e[i]==diag(LO[i],-LO[i])                   : True
Gamma17(x)^2 == (x.eta17.x) 1                                           : True
anticommutant of G17: dimension 1 (numerical rank 255 of 2048x256 system): True
G17[8] anticommutes with all G17[A], G17[8]^2 == -1                     : True
null k: k.eta17.k == 0, kslash^2 == 0, 8 rational solutions of kslash w = 0: True
Psi = w cos(k.x) is a REAL solution of G17^A d_A Psi = 0                : True
p.eta71.p == -m^2 and P32^2 == -m^2                                     : True
Psi32 (32 arbitrary real w) solves (G71R32^A d_A - m) Psi = 0           : True
Psi32 obeys the Klein-Gordon equation (box17 + m^2) Psi = 0             : True
(G71R32^A d_A)^2 == -box17 on 32 arbitrary functions                    : True
```

tools/ref_dirac16.py (independent checks of sections 7-9; exit code 0)

```
Gamma17 real, entries 0,+-1 : True
{Gamma_A,Gamma_B} = 2 eta17_AB : True
Gamma0 symmetric, Gamma_i antisymmetric : True
Gamma^A dagger = Gamma0 Gamma^A Gamma0 : True
alpha^i, beta real symmetric : True
{alpha_i,alpha_j}=2delta, {alpha_i,beta}=0, beta^2=1 : True
block forms alpha^i = [[0,tauO_i],[-tauO_i,0]], beta = [[0,1],[1,0]] : True
omega = Gamma17[8] = [[0,-1],[1,0]], omega^2 = -1 : True
(i Dslash + m)(i Dslash - m) psi = -(box + m^2) psi : True
Gamma0 (i d0 - h) psi = (i Dslash - m) psi : True
```

```
h(p) = [[0,M],[M^T,0]] : True
h(p)^2 = E^2 1 : True
M M^T = M^T M = E^2 1_8 : True
h(p) u_s = E u_s,  h(-p) v_s = -E v_s : True
(pslash - m) u_s = 0, (pslash + m) v_s = 0 : True
u_s^T u_r = 2E delta, v_s^T v_r = 2E delta : True
u_s(p)^T v_r(p~) = 0 : True
sum_s u_s ubar_s = pslash + m, sum_s v_s vbar_s = pslash - m : True
completeness sum u u^T + sum v(p~) v(p~)^T = 2E 1_16 : True
U_p orthogonal and U_p^T h(p) U_p = diag(E 1_8, -E 1_8) : True
charge conjugation: (pslash + m) omega u_s = 0 : True
omega u_s = sum_r v_r (-M^T/E)_rs : True
S Gamma_K S^-1 = sum_B Lambda_BK Gamma_B (boost) : True
Lorentz covariance: (pslash(Lambda p) - m) S u_s(p) = 0 : True
real matrices anticommuting with all Gamma17[A]: dimension 1 (rank 255) : True
omega anticommutes with all Gamma17[A] : True
chi = i omega: Hermitian, chi^2 = 1, {chi,Gamma_A} = 0 : True
[chi, Sigma_AB] = 0 (chirality is Lorentz invariant) : True
P_L, P_R not real : True
P_L+P_R = 1, P_L^2 = P_L, P_R^2 = P_R, P_L P_R = 0, rank 8 each : True
Gamma_A P_L = P_R Gamma_A,  P_L^dagger Gamma0 = Gamma0 P_R : True
E_L, E_R isometries: E^dag E = 1_8, E_L^dag E_R = 0, E_L E_L^dag = P_L, E_R E_R^dag = P_R : True
chi E_L = -E_L, chi E_R = E_R, W unitary : True
W^dag h(p) W = [[-sigma.p, m],[m, sigma.p]] : True
sigma_i Hermitian, {sigma_i,sigma_j} = 2 delta : True
chiral blocks: E_L^dag beta E_L = 0, E_L^dag beta E_R = 1, E_L^dag alpha E_L = -sigma, E_R^dag
    ...alpha E_R = sigma, E_L^dag alpha E_R = 0 : True
E_L^dag Gamma0 (i Dslash - m) psi = (i d0 + tauO.d) phi_L - m phi_R : True
E_R^dag Gamma0 (i Dslash - m) psi = (i d0 - tauO.d) phi_R - m phi_L : True
(i d0 - tauO.d)(i d0 + tauO.d) f = -box f (8 components) : True
psi = E_L phi_L + E_R phi_R : True
chiral form of the bilinears (beta i chi, chi, alpha chi) : True
{psi_a(t,x), psi_b^dag(t,y)} = delta_ab (1/V) sum_q e^{iq(x-y)} (5 momenta, abstract mode
    ...algebra) : True
{psi_a(t,x), psi_b(t,y)} = 0 : True
the three U_k solve the Dirac equation (m = 1) : True
Grassmann components anticommute: psi_a psi_b + psi_b psi_a = 0, psi_a^2 = 0 : True
current conservation d_A (psibar Gamma^A psi) = 0 (Grassmann field) : True
dL_int/dpsibar_a = -[gS(psibar psi) psi + gP (psibar i chi psi) i chi psi + gV (psibar G_A psi)
    ...G^A psi + gA (psibar G_A chi psi) G^A chi psi]_a : True
h_KS Hermitian for real S, P, V_A, A_A : True
Gamma0 [pslash(eps,p) - m - S - i chi P - Gamma^A (V_A + chi A_A)] = eps - h_KS(p)  (plane wave
    ...e^{-i(eps t - p.x)}) : True
W^dag h_KS W = [[V^L_0 - sigma.(p+V^L), M],[M^*, V^R_0 + sigma.(p+V^R)]], V^L = V - A, V^R = V
    ...+ A, M = m + S + iP : True
Hartree potentials: dE_H/dphi*_a = (v_KS phi)_a with S = gS rhoS, P = gP rhoP, V_A = gV j_A,
    ...A_A = gA j5_A : True
densities real: Gamma0 O Hermitian for O = 1, i chi, Gamma^A, Gamma^A chi : True
chiral densities: rhoS = kappa + kappa^*, rhoP = i(kappa - kappa^*), j = jL + jR, j5 = jR - jL
    ...: True
E_H = gS/2 (k+k*)^2 - gP/2 (k-k*)^2 + (gV+gA)/2 (jL.jL + jR.jR) + (gV-gA) jL.jR : True
V^L_A = V_A - A_A, V^R_A = V_A + A_A (Hartree, chiral currents) : True
M - m = dE_H/dkappa = gS(k+k*) - gP(k-k*) = S + iP : True
dE_H/dphi_L* = (V^L_0 - sigma_i V^L_i) phi_L + (M - m) phi_R ; dE_H/dphi_R* = (V^R_0 + sigma_i
    ...V^R_i) phi_R + (M - m)^* phi_L : True
U_theta = cos(theta) - sin(theta) omega is real orthogonal; U E_L = e^{-i theta} E_L, U E_R =
    ...e^{i theta} E_R : True
U^dag beta U = beta (cos 2theta + i sin 2theta chi);  vector and axial currents invariant : True
E_H(e^{i theta chi} psi) - E_H(psi) = (gS-gP)/2 [(e^{4i theta}-1) k^2 + (e^{-4i theta}-1) k*^2]
    ...->  U(1)_A symmetric iff gP = gS : True
256 products Gamma_K are orthogonal: Gamma_K^-1 = Gamma_K^T : True
tr(Gamma_K^-1 Gamma_L) = 16 delta_KL (256 x 256) : True
Fierz completeness: sum_K (Gamma_K)_ab (Gamma_K^-1)_cd = 16 delta_ad delta_cb : True
delta E_x = tr[Sigma_x delta R] (random R, dR), Sigma_x = -gS R - gP (i chi) R (i chi) - gV G_A
    ...R G^A - gA G_A chi R G^A chi : True
```

```
E_x = (1/2) tr[Sigma_x R] : True
E_H = (gS/2)(tr R)^2 + ... = sum_c (g_c/2) tr(O_c R) tr(O^c R) and dE_H = tr[Sigma_H dR] : True
Fierz: tr[O1 R O2 R] = (1/16) sum_K tr(Gamma_K^-1 R) tr(O1 Gamma_K O2 R) for all channel
    ...vertices : True
uniform R: tr R = rho_S, tr(Gamma^0 R) = rho : True
(1/2) sum_s [u_s(p) ubar_s(p) + u_s(-p) ubar_s(-p)]/(2E) = (E Gamma_0 + m)/(2E) : True
E_x(uniform) = -gS/32 (rS^2+rho^2) + gP/32 (rS^2-rho^2) - gV/32 (8rS^2-6rho^2) + gA/32
    ...(8rS^2+6rho^2) : True
Sigma_x(uniform) = S_x + V_x Gamma_0, S_x = rS/16(-gS+gP-8gV+8gA), V_x = rho/16(-gS-gP+6gV+6gA)
    ...: True
S_x = dE_x/drho_S, V_x = dE_x/drho : True
uniform Hartree-Fock = Hartree with G_S = gS + (-gS+gP-8gV+8gA)/16, G_V = gV +
    ...(-gS-gP+6gV+6gA)/16 : True
uniform matter: Hartree terms of P and A vanish (rho_P = 0, j5 = 0, j^i = 0) : True
Wick (3 particles, 16 modes): <:(psibar O psi)(psibar O' psi):> = tr(O R) tr(O' R) - tr(O R O'
    ...R) for all 18 vertex pairs : True
c7 = 8 |S^6|/(2pi)^7 = 1/(15 pi^4) and rho = c7 kF^7/7 = kF^7/(105 pi^4) : True
NUM Hartree saturation kF0 : 0.75663615070149289469
NUM Hartree rho0 : 0.00001388100213231687719 7
NUM Hartree m* : 0.34809734014624326514
NUM Hartree rhoS : 6.5190265985375673486e-6
NUM Hartree V0 : 0.13881002132316877197
NUM Hartree E/A - m : -0.028321427964738575854
NUM Hartree eps_F : 0.97167857203526142415
Hartree gap: m* - m - gS rhoS(m*) = 0 : True
m* is a stationary point of eps(m*) = ekin(m*) - (m*-m)^2/(2gS) + gV rho^2/2 : True
total energy = sum eps_k - E_H (double counting) = ekin - (m*-m) rhoS + E_H : True
Hugenholtz-Van Hove at saturation: E/A = eps_F = d eps/d rho : True
d(eps/rho)/dkF = 0 at kF0 (minimum: E/A(0.95 kF0), E/A(1.05 kF0) larger) : True
Hugenholtz-Van Hove mu = d eps/d rho = eps_F at kF = 1 (not saturated) : True
NUM Hartree kF=1: m* : 0.080964430277606126329
NUM Hartree kF=1: E/A - m : 0.41086484863433217115
NUM Hartree kF=1: eps_F : 1.9809848613081542876
NUM HF couplings G_S, G_V : -98750.0, 20000.0
HF self-consistency with the exchange self-energy: m* - m = gS rhoS + S_x, V0 = gV rho + V_x :
    ...True
HF energy: ekin - (m*-m) rhoS + E_H + E_x = eps(G_S, G_V) : True
HF Hugenholtz-Van Hove: d eps/d rho = eps_F at kF = 3/4 : True
NUM HF kF=3/4: m* : 0.36643630198154392124
NUM HF kF=3/4: V0 : 0.2610177547316155257
NUM HF kF=3/4: E/A - m : 0.039146338016119861526
NUM Hartree kF=3/4: m* : 0.36297394764803019822
NUM Hartree kF=3/4: E/A - m : -0.028065375846872960879
J(0) = Lambda^6/(720 pi^4), g_c = -1/(8 J(0)) = -90 pi^4/Lambda^6 : True
NUM NJL g_c : -8766.8181930602193513
NUM NJL g = 6/5 g_c : -10520.181831672263222
NUM NJL |M| (m = 0) : 0.55369385182924169122
NUM NJL kappa = -4 M J(|M|) (M real) : -0.00002631579285836487 42
NUM NJL E(|M|) - E(0) : -1.3102449629010022473e-6
NJL m = 0: gap 1 + 8 g J(|M|) = 0 solved, M = 2 g kappa^*, stationary and lower than M = 0 :
    ...True
NJL m = 0: E(|M| e^{i theta}) independent of theta (degenerate chiral vacua) : True
NUM NJL M (m = 1/100) : 0.58531284674622051123
NJL m = 1/100: M = m - 8 g J(M) M real, stationary; E(M e^{i theta}) > E(M) for theta != 0 :
    ...True
sea at one momentum: eigenvalues -E (x8), +E (x8); kappa_p = tr[(Pi_-)_{RL}] = -4 M^*/E;
    ...tr[Pi_- h_0] = -8E + 8 Re[(M-m) M^*]/E : True
boolean checks: 90   failed: 0
VERDICT: ALL CHECKS PASSED
```

tools/ref_gravity.py (independent checks of sections 11-14; exit code 0)

```
cosmological metric: sectional curvatures R^{AB}_{AB} = -(Ha'+Ha^2), -(Hb'+Hb^2), -Ha^2, -Hb^2,
    ...-Ha Hb : True
```

```
cosmological metric: Riemann R^{cd}_{ab} non-zero only for {c,d} = {a,b} : True
G^0_0 = 3Ha^2 + 12 Ha Hb + 6 Hb^2 : True
G^1_1 = 2Ha' + 3Ha^2 + 4Hb' + 10Hb^2 + 8HaHb : True
G^4_4 = 3Ha' + 6Ha^2 + 3Hb' + 6Hb^2 + 9HaHb : True
scalar curvature R = -2(3Ha' + 6Ha^2 + 4Hb' + 10Hb^2 + 12 HaHb) : True
E^(1) = Einstein tensor G (cosmological metric) : True
L_1 = R : True
E^(0) = -delta/2 : True
trace identity E^(k)mu_mu = -(8-2k)/2 L_k, k = 0..4 : True
E^(4) = 0 identically in 8 dimensions, but L_4 (Euler density) is not 0 : True
E^(k) diagonal: E^(k)mu_nu = 0 for mu != nu, and E^(k)1_1 = E^(k)2_2 = E^(k)3_3, E^(k)4_4 = ...
    ...= E^(k)7_7 : True
E^(2)0_0 = -12 Hb (4Ha^3 + 18Ha^2Hb + 12HaHb^2 + Hb^3) : True
E^(3)0_0 = 360 Ha^2 Hb^3 (4Ha + 3Hb) : True
E^(2) = Lanczos tensor 2(R R_mn - 2R_ma R^a_n - 2R^ab R_manb + R_m^abc R_nabc) - g_mn L_2/2,
    ...and L_2 = R^2 - 4 Ric.Ric + Riem.Riem (cosmological metric) : True
closed forms : {(1, 0): 3*H_a**2 + 12*H_a*H_b + 6*H_b**2, (1, 1): 3*H_a**2 + 8*H_a*H_b +
    ...10*H_b**2 + 2*dH_a + 4*dH_b, (1, 4): 6*H_a**2 + 9*H_a*H_b + 6*H_b**2 + 3*dH_a + 3*dH_b,
    ...(2, 0): -48*H_a**3*H_b - 216*H_a**2*H_b**2 - 144*H_a*H_b**3 - 12*H_b**4, (2, 1):
    ...-32*H_a**3*H_b - 136*H_a**2*H_b**2 - 16*H_a**2*dH_b - 192*H_a*H_b**3 - 32*H_a*H_b*dH_a -
    ...96*H_a*H_b*dH_b - 60*H_b**4 - 48*H_b**2*dH_a - 48*H_b**2*dH_b, (2, 4): -12*H_a**4 -
    ...108*H_a**3*H_b - 180*H_a**2*H_b**2 - 12*H_a**2*dH_a - 36*H_a**2*dH_b - 108*H_a*H_b**3 -
    ...72*H_a*H_b*dH_a - 72*H_a*H_b*dH_b - 12*H_b**4 - 36*H_b**2*dH_a - 12*H_b**2*dH_b, (3, 0):
    ...1440*H_a**3*H_b**3 + 1080*H_a**2*H_b**4, (3, 1): 576*H_a**3*H_b**3 + 1368*H_a**2*H_b**4
    ...+ 864*H_a**2*H_b**2*dH_b + 576*H_a*H_b**5 + 576*H_a*H_b**3*dH_a + 576*H_a*H_b**3*dH_b +
    ...144*H_b**4*dH_a, (3, 4): 648*H_a**4*H_b**2 + 1224*H_a**3*H_b**3 + 432*H_a**3*H_b*dH_b +
    ...648*H_a**2*H_b**4 + 648*H_a**2*H_b**2*dH_a + 648*H_a**2*H_b**2*dH_b +
    ...432*H_a*H_b**3*dH_a}
divergence-free E^(k), k = 1, 2, 3 (cosmological metric, arbitrary a, b) : True
variation of sqrt|g| L_k (lapse N, a, b): EL_N = -2 a^3 b^4 E^(k)0_0, EL_a = -6 a^2 b^4
    ...E^(k)1_1, EL_b = -8 a^3 b^3 E^(k)4_4, k = 1..4 : True
inhomogeneous metric: E^(3)0_4 at the point (exact) :
    ...353743050254866364332989413672793316453216419840000/24577736888913221711303629009057562607
    ...63178144678501
inhomogeneous metric (N, A, B of x0 and x4): E^(1) = G at the point : True
inhomogeneous metric: nabla_mu E^(k)mu_nu = 0 exactly at x0 = 1/2, x4 = 1/3, k = 1, 2, 3 : True
inhomogeneous metric: E^(k)_mu nu symmetric, k = 1, 2, 3 : True
pp-wave: E^(1)_mu nu = +(1/2) (d2^2 + d3^2) F l_mu l_nu, l = dx0 - dx1 : True
pp-wave: E^(2) = E^(3) = 0 for every profile F : True
Kasner E^(k) components x0^(2k) (00, 11, 44) : {1: [3*(p**2 + 4*p*q + 2*q**2), 3*p**2 + 8*p*q -
    ...2*p + 10*q**2 - 4*q, 3*(2*p**2 + 3*p*q - p + 2*q**2 - q)], 2: [-12*q*(4*p**3 + 18*p**2*q
    ...+ 12*p*q**2 + q**3), -4*q*(8*p**3 + 34*p**2*q - 12*p**2 + 48*p*q**2 - 36*p*q + 15*q**3 -
    ...12*q**2), -12*(p**4 + 9*p**3*q - p**3 + 15*p**2*q**2 - 9*p**2*q + 9*p*q**3 - 9*p*q**2 +
    ...q**4 - q**3)], 3: [360*p**2*q**3*(4*p + 3*q), 72*p*q**3*(8*p**2 + 19*p*q - 20*p + 8*q**2
    ...- 10*q), 72*p**2*q**2*(9*p**2 + 17*p*q - 15*p + 9*q**2 - 15*q)]}
Einstein Kasner: p = (1+-2 sqrt2)/7, q = (2-+3 sqrt2)/14 solve E^(1) = 0; 3p+4q = 1, 3p^2+4q^2
    ...= 1 : True
Gauss-Bonnet Kasner cubic (64 x e4 at q = (3-3p)/4) : -203*p**3 + 81*p**2 + 351*p + 27
Gauss-Bonnet Kasner exponents p (three real roots) : (-1.083501787612105742423976294 73,
    ...-0.078631058418927766227861052 1099, 1.561147624356156661361196952 75)
Gauss-Bonnet Kasner: the three (p, (3-3p)/4) solve E^(2) = 0, sum of the 7 exponents = 3 : True
cubic Lovelock Kasner: (p, q) = (-15/7, 20/7) solves E^(3) = 0, 3p + 4q = 5; degenerate
    ...families q = 0 and p = 0 : True
Einstein Kasner is not a cubic-Lovelock vacuum (E^(3) != 0) : True
8D de Sitter (a = b = e^{H x0}): E^(1) = 21 H^2, E^(2) = -420 H^4, E^(3) = 2520 H^6 (times
    ...delta) : True
4D de Sitter on a static T^4 (a = e^{H x0}, b = b0): E^(1) = diag(3,3,6) H^2, E^(2) =
    ...diag(0,0,-12) H^4, E^(3) = 0 : True
static T^4 (Hb = 0): E^(1) = (3Ha^2, 2Ha'+3Ha^2, 3Ha'+6Ha^2), E^(2) = (0, 0, -12 Ha^2
    ...(Ha'+Ha^2)), E^(3) = 0 : True
exponential expansion a = e^{H x0}, b = e^{q H x0}: E^(k) (00, 11, 44) = closed forms with Ha =
    ...H, Hb = q H, Ha' = Hb' = 0 (k = 1, 2, 3) : True
exponential expansion: 3 Ha (eq00 - eq11) + 4 Hb (eq00 - eq44) = 0 identically (the three
    ...vacuum equations have rank 2) : True
exponential vacuum solutions: one solution (w2, w3) for given w1, Lambda, q, satisfying all
```

```
    ...three equations : True
eternal inflation a = e^{H x0} with deflation b = e^{-H x0}: w2 = Lambda/4 - w1/2, w3 = w1/24 -
    ...Lambda/36 (Lambda = 0: w2/w1 = -1/2, w3/w1 = 1/24) : True
NUM exponential vacuum solution (w2, w3) for general q : ((-2*Lambda*q - 3*Lambda + 12*q**3*w1
    ...+ 27*q**2*w1 + 22*q*w1 + 9*w1)/(4*q*(6*q**4 + 36*q**3 + 66*q**2 + 51*q + 16)),
    ...(-3*Lambda*q**2 - 6*Lambda*q - Lambda + 15*q**4*w1 + 36*q**3*w1 + 33*q**2*w1 + 18*q*w1 +
    ...3*w1)/(72*q**3*(6*q**4 + 36*q**3 + 66*q**2 + 51*q + 16)))
NUM eq00, eq11, eq44 for q = -1 : (-Lambda - 3*w1 - 36*w2 - 360*w3, -Lambda + 5*w1 + 28*w2 +
    ...216*w3, -Lambda + 3*w1 + 12*w2 + 72*w3)
cosmological metric: eta e e = g : True
cosmological metric: {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
cosmological metric: Gamma_mu real : True
cosmological metric: omega_mu AB = -omega_mu BA : True
cosmological metric: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu =
    ...0 : True
cosmological metric: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
cosmological metric: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu
    ...nu Gamma_l + [Omega_mu, Gamma_nu] = 0 : True
cosmological metric: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 :
    ...True
cosmological metric: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B : True
cosmological metric: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 : True
cosmological metric: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
cosmological metric: Gamma^mu Omega_mu = (3a'/(2a) + 2b'/b) Gamma17[0] : True
cosmological metric: Omega_i = (a'/2) Gamma17[i] Gamma17[0] (i = 1..3), Omega_k = (b'/2)
    ...Gamma17[k] Gamma17[0] (k = 4..7), Omega_0 = 0 : True
pp-wave: eta e e = g : True
pp-wave: {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
pp-wave: Gamma_mu real : True
pp-wave: omega_mu AB = -omega_mu BA : True
pp-wave: vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 : True
pp-wave: Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
pp-wave: total covariant derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l
    ...+ [Omega_mu, Gamma_nu] = 0 : True
pp-wave: constant frame matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 : True
pp-wave: spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A Gamma^B : True
pp-wave: Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0 : True
pp-wave: (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
inhomogeneous metric N, A, B (x0, x4): eta e e = g : True
inhomogeneous metric N, A, B (x0, x4): {Gamma_mu, Gamma_nu} = 2 g_mu nu : True
inhomogeneous metric N, A, B (x0, x4): Gamma_mu real : True
inhomogeneous metric N, A, B (x0, x4): omega_mu AB = -omega_mu BA : True
inhomogeneous metric N, A, B (x0, x4): vielbein postulate d_mu e^A_nu - chr^l_mu nu e^A_l +
    ...omega_mu^A_B e^B_nu = 0 : True
inhomogeneous metric N, A, B (x0, x4): Omega_mu real, Gamma17[0] Omega_mu^T Gamma17[0] =
    ...-Omega_mu : True
inhomogeneous metric N, A, B (x0, x4): total covariant derivative of the frame matrices: d_mu
    ...Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu, Gamma_nu] = 0 : True
inhomogeneous metric N, A, B (x0, x4): constant frame matrices: omega_mu^A_B Gamma^B +
    ...[Omega_mu, Gamma^A] = 0 : True
inhomogeneous metric N, A, B (x0, x4): spin curvature F_mu nu = (1/4) R_AB mu nu Gamma^A
    ...Gamma^B : True
inhomogeneous metric N, A, B (x0, x4): Bianchi identity d_l F_mn + [Omega_l, F_mn] + cyclic = 0
    ...: True
inhomogeneous metric N, A, B (x0, x4): (1/2) Gamma^mu Gamma^nu F_mu nu = -(R/4) 1_16 : True
Cartan's first structure equation de^A + omega^A_B ^ e^B = 0 (torsion-free) for the
    ...cosmological frame : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): eta e e = g : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): {Gamma_mu, Gamma_nu} =
    ...2 g_mu nu : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Gamma_mu real : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): omega_mu AB =
    ...-omega_mu BA : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): vielbein postulate
    ...d_mu e^A_nu - chr^l_mu nu e^A_l + omega_mu^A_B e^B_nu = 0 : True
```

```
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Omega_mu real,
    ...Gamma17[0] Omega_mu^T Gamma17[0] = -Omega_mu : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): total covariant
    ...derivative of the frame matrices: d_mu Gamma_nu - chr^l_mu nu Gamma_l + [Omega_mu,
    ...Gamma_nu] = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): constant frame
    ...matrices: omega_mu^A_B Gamma^B + [Omega_mu, Gamma^A] = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): spin curvature F_mu nu
    ...= (1/4) R_AB mu nu Gamma^A Gamma^B : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): Bianchi identity d_l
    ...F_mn + [Omega_l, F_mn] + cyclic = 0 : True
static radial metric f(r) dx0^2 - h(r) dr^2 - r^2 (dx2^2 + ... + dx7^2): (1/2) Gamma^mu
    ...Gamma^nu F_mu nu = -(R/4) 1_16 : True
Cartan's first structure equation for the static radial metric : True
structure coefficients: de^A = (1/2) C^A_BC e^B ^ e^C and omega_C,AB (form index C) =
    ...-(1/2)(C_ACB + C_BAC - C_CBA) (static radial metric) : True
static radial metric: Omega_0 = (f'/(4 sqrt(f h))) Gamma17[1] Gamma17[0], Omega_1 = 0, Omega_k
    ...= (1/(2 sqrt h)) Gamma17[1] Gamma17[k] (k = 2..7) : True
static radial metric: Gamma^mu Omega_mu = (1/sqrt h)(f'/(4f) + 3/r) Gamma17up[1], so (i
    ...Gamma^mu D_mu - m) psi = [i Gamma17up[0] d0/sqrt f + i Gamma17up[1] (d_r + f'/(4f) +
    ...3/r)/sqrt h + sum_k i Gamma17up[k] d_k/r - m] psi : True
static radial metric: psi = f^(-1/4) r^(-3) chi removes the spin connection: Gamma^mu D_mu psi
    ...= f^(-1/4) r^(-3) Gamma^mu d_mu chi : True
U(1) gauge covariance: Gamma^mu (D_mu - i q A'_mu)(e^{i q alpha} psi) = e^{i q alpha} Gamma^mu
    ...(D_mu - i q A_mu) psi with A' = A + d alpha : True
current identity with the gauge field: d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu
    ...psibar) Gamma^mu psi + psibar Gamma^mu D_mu psi], D including -i q A_mu : True
current identity d_mu(sqrt|g| psibar Gamma^mu psi) = sqrt|g| [(D_mu psibar) Gamma^mu psi +
    ...psibar Gamma^mu D_mu psi] : True
Lichnerowicz: (Gamma^mu D_mu)^2 psi = (g^mu nu nabla_mu D_nu - R/4) psi : True
rescaling psi = (a^3 b^4)^(-1/2) chi removes the spin connection: Gamma^mu D_mu psi = (a^3
    ...b^4)^(-1/2) Gamma^mu d_mu chi : True
mode equation: (i Gamma^mu D_mu - m) psi = 0  <=>  i dc/dx0 = h(x0) c, h = alpha.(p/a, q/b) +
    ...beta m : True
conformal weight 7/2: Gamma^mu D_mu (C^(-7/2) f) = C^(-9/2) Gamma^A d_A f for g = C(x0)^2 eta :
    ...True
2x2 reduction: A = alpha.n_a, B = alpha.n_b, beta: squares 1, mutually anticommuting; Z = A B
    ...beta: Z^2 = -1, Z^T = -Z, commutes with A, B, beta : True
Kasner (Einstein branch), massless: |beta|^2 for (|p|, |q|) = (1,1), (2,1), (1,3) :
    ...(np.float64(0.01074934616777871), np.float64(0.00013294108220674155),
    ...np.float64(0.06744202228393215))
Kasner scaling variable K = |p| lambda^(1-p), lambda = (|q|/|p|)^(1/(q-p)) : (1.0,
    ...3.11828865465698, 0.4946322370571032)
Kasner: |alpha|^2 + |beta|^2 = 1 (to 1e-8) : True
Kasner scaling: (|p|, |q|) = (1.430..., 2.5) has K = 1 and the same |beta|^2 as (1, 1) (to
    ...1e-6) : True
Kasner, m = 0.3, |p| = 1, |q| = 1.5, x0: 0.05 -> 40: |beta|^2 of the 16x16 evolution (average),
    ...of the sector Z = +i (+m sigma3) and of the sector Z = -i (-m sigma3) :
    ...(np.float64(0.027052647156543617), np.float64(0.02528659275098409),
    ...np.float64(0.028818701562187667))
16x16 evolution = 4 copies of the 2x2 evolution with +m sigma3 and 4 copies with -m sigma3:
    ...average agrees (to 1e-8), unitarity; the two sectors differ : True
isotropic expansion, massless: no pair creation (|beta|^2 < 1e-20) : True
far future (de Sitter, H = 1): |beta|^2 for m/H = 0.25, 0.5, 1 :
    ...(np.float64(0.1721029191835808), np.float64(0.04142388447252214),
    ...np.float64(0.001863977463333981))
de Sitter: |beta|^2 = 1/(exp(2 pi m/H) + 1) (Fermi-Dirac at the Gibbons-Hawking temperature
    ...H/2pi), to 1e-7 : True
integrator benchmark: Landau-Zener |beta|^2 = exp(-pi kappa^2) (to 1e-6) : True
boolean checks: 99   failed: 0
VERDICT: ALL CHECKS PASSED
```

tools/ref_ks_gravity.py (independent checks of sections 15-16; exit code 0)

```
256 products in 9 ranks: 1, 8, 28, 56, 70, 56, 28, 8, 1 : True
every bilinear psibar Theta^K psi is real: Gamma17[0] Theta^K Hermitian (Theta^K =
    ...i^((r(r-1)/2) mod 2) Gamma^K) : True
Theta_K Theta^K = 1_16 and tr(Theta_K Theta^J) = 16 delta_KJ : True
Fierz completeness: sum_K (Theta_K)_cd (Theta^K)_ab = 16 delta_ad delta_bc : True
Hodge duality: Theta^K omega17 = +- Theta^(complement of K) (up to a phase) : True
stage-2 channels: I_0 = (psibar psi)^2, I_1 = (psibar Gamma^A psi)(psibar Gamma_A psi), I_8 =
    ...-(psibar i chi psi)^2, I_7 = -(axial) : True
local Lorentz invariance: I_r(S psi, psibar S^-1) = I_r(psi, psibar), r = 0..8 (random finite
    ...boost-rotation) : True
Fierz matrix F_rs = (1/256) sum_{K in r} tr(Theta_K Theta_J Theta^K Theta^J), independent of
    ...the representative J of rank s : True
Fierz matrix closed form F_rs = (1/16) sum_l (-1)^(rs-l) C(s,l) C(8-s,r-l) : True
F^2 = 1 (the Fierz transformation is an involution) : True
eigenvalues of F: number of +1, number of -1 : (5, 4)
Fierz matrix (rows r, columns s) times 16 : [[1, 1, 1, 1, 1, 1, 1, 1, 1], [8, -6, 4, -2, 0, 2,
    ...-4, 6, -8], [28, 14, 4, -2, -4, -2, 4, 14, 28], [56, -14, -4, 6, 0, -6, 4, 14, -56],
    ...[70, 0, -10, 0, 6, 0, -10, 0, 70], [56, 14, -4, -6, 0, 6, 4, -14, -56], [28, -14, 4, 2,
    ...-4, 2, 4, -14, 28], [8, 6, 4, 2, 0, -2, -4, -6, -8], [1, -1, 1, -1, 1, -1, 1, -1, 1]]
Fierz identity for commuting spinors: sum_{K in r} (a Theta_K b)(c Theta^K d) = sum_s F_rs
    ...sum_{J in s} (a Theta_J d)(c Theta^J b) : True
Hartree-Fock energy E_H + E_x = sum_s (G_s/2) D_s with G = (1 - F^T) g (random density matrix,
    ...random couplings) : True
uniform matter R = (rho_S + rho Gamma17[0])/16: only D_0 = rho_S^2, D_1 = rho^2; with g_0 = gS,
    ...g_1 = gV, g_7 = -gA, g_8 = -gP the exchange energy is that of stage 2 (section 8.7) :
    ...True
uniform density matrix: D_0 = rho_S^2, D_1 = rho^2, D_2..D_8 = 0 : True
stage-2 couplings gS = -100000, gV = 10000: G_0 = -98750, G_1 = 20000 (the HF couplings of
    ...section 8.7) : True
effective couplings G_0, G_1 in terms of all nine g_r : (15*g0/16 - g1/2 - 7*g2/4 - 7*g3/2 -
    ...35*g4/8 - 7*g5/2 - 7*g6/4 - g7/2 - g8/16, -g0/16 + 11*g1/8 - 7*g2/8 + 7*g3/8 - 7*g5/8 +
    ...7*g6/8 - 3*g7/8 + g8/16)
example g = (-100000, 10000, 3000, 0, ..., 0): G_0, G_1 : (-104000, 17375)
nonlinear field equation: dL_int/d psibar_a = -sum_r g_r sum_K (psibar Theta_K psi)(Theta^K
    ...psi)_a : True
the chirality chi17 commutes with every generator Sigma^AB, hence with every spin connection
    ...Omega_mu : True
curved chiral equations: E_L^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) +
    ...tauO.d/(a,b)) phi_L - m phi_R : True
curved chiral equations: E_R^dag Gamma0 (i Gamma^mu D_mu - m) psi = (i d0 + i(3Ha/2 + 2Hb) -
    ...tauO.d/(a,b)) phi_R - m phi_L : True
flat limit a = b = 1, kF = kF0 (stage-2 saturation): m* = 0.3480973401..., E/A - m =
    ...-0.0283214279... : True
flat limit: isotropic pressure pa = pb = rho eps_F - eps (Gibbs-Duhem, eps_F = V0 + sqrt(kF^2 +
    ...m*^2)) : True
NUM flat: m*, rho, eps, p : (0.3480973401462447, np.float64(1.3881002132316779e-05),
    ...np.float64(1.3487872330347983e-05), np.float64(-1.588186776101813e-20))
energy-momentum conservation eps' + 3Ha(eps + pa) + 4Hb(eps + pb) = 0 for the KS matter
    ...(arbitrary a, b; relative residual) : True
NUM conservation residual / eps' : -5.166108748755716e-12
primordial (Einstein-Kasner background, comoving kF = kF0 at x0 = 1): m*(x0) for x0 = 1e-4 ...
    ...10 : (0.003824179379430487, 0.010783989711922532, 0.030238519612042748,
    ...0.08844464116180238, 0.3480973401462447, 0.8917637493589158)
primordial: rho(x0) x0 = kF0^7/(105 pi^4) (density grows like 1/x0) : True
primordial: m* decreases toward 0 as x0 -> 0 (blue-shifted ordinary momenta, attractive scalar
    ...coupling) : True
primordial: anisotropic pressure pa/pb at x0 = 1e-3, 1e-1, 10 :
    ...(np.float64(1.0868869169150372), np.float64(1.6461737747098435),
    ...np.float64(-0.4036333858521348))
modern (static auxiliary space b = 1): m*(a) for a = 1, 2, 4, 10, 100 : (0.3480973401462447,
    ...0.853540809048541, 0.9807410717586909, 0.9987576007278518, 0.9999987563024887)
modern: eps a^3 -> constant and pa/eps -> 0 (dirac16complex behaves as pressureless matter in
    ...ordinary space) : True
modern: pb/eps at a = 100 (stress along the auxiliary space, balanced by the stabilizing
```

```
    ...stress) : 0.049773512087215956
far future (de Sitter a = e^{0.1 x0}, b = 1): m*(x0) for x0 = 0, 10, 20, 40, 80 -> m = 1 :
    ...(0.3480973401462447, 0.9395796910293632, 0.996923581335845, 0.9999923586141617,
    ...0.9999999999530483)
far future: rho = rho0 e^{-3 H x0} and 1 - m* -> 0 like rho : True
the 00 closed forms of E^(1), E^(2), E^(3) (pair-set algorithm, tools/lovelock71.py) agree with
    ...the closed forms of section 12.6 of 7+1.nb : True
Einstein-Lovelock equations are linear in (Ha', Hb') (quasi-linear second-order system) : True
Einstein + KS matter (kappa = 1000, Lambda = 0, Ha(0) = 0.05): Hb(0) from the constraint, (a,
    ...b, Ha, Hb) at x0 = 20, max relative constraint violation : (0.009143712444558836,
    ...(np.float64(1.5817386798469366), np.float64(1.1975585618423046),
    ...np.float64(0.011877999367582402), np.float64(0.007419107766685833)),
    ...np.float64(8.415965253866209e-12))
Einstein-Lovelock (alpha2 = 20, alpha3 = 500) + KS matter: Hb(0), (a, b, Ha, Hb) at x0 = 20,
    ...max relative constraint violation : (0.014980703942304356,
    ...(np.float64(1.4773514780943342), np.float64(1.276084053398673),
    ...np.float64(0.010175914492223711), np.float64(0.008488764693542736)),
    ...np.float64(4.387597609521105e-12))
the constraint Sum alpha_k E^(k)0_0 - Lambda = kappa eps is preserved by the evolution
    ...(relative violation < 1e-7) in both cases : True
boolean checks: 29   failed: 0
VERDICT: ALL CHECKS PASSED
```

tools/ref_sm.py (independent checks of sections 17-19; exit code 0)

```
anomaly cancellation in each generation (with nu_R): SU(3)^2U(1), SU(2)^2U(1), U(1)^3,
    ...grav^2U(1), SU(3)^3 all 0 : True
Witten SU(2) anomaly: number of doublets per generation (3 colours of Q + 1 L) = 4, even : True
electric charges Q = T3 + Y: Q -> (2/3, -1/3), u_R 2/3, d_R -1/3, L -> (0, -1), e_R -1, nu_R 0
    ...: True
effective number of relativistic degrees of freedom g* = 28 + (7/8) 90 = 106.75 (106.75 + 5.25
    ...if nu_R were thermal) : True
gauge-boson mass matrix from |D<Phi>|^2: eigenvalues g^2v^2/4 (twice, W+-), (g^2+g'^2)v^2/4
    ...(Z), 0 (photon) : True
NUM v = (sqrt2 G_F)^(-1/2) [GeV], g, g', sin^2 theta_W (on shell), e, 1/alpha(tree), lambda =
    ...m_H^2/(2v^2) : (246.21964023926205, 0.6528252573344827, 0.34994623401223707,
    ...0.22320949284344327, 0.3084276316025824, 132.10014858375337, 0.1292805756612353)
tree-level relations: m_W = g v/2, m_Z = sqrt(g^2+g'^2) v/2, rho = m_W^2/(m_Z^2 cos^2) = 1,
    ...m_H^2 = 2 lambda v^2 : True
CKM unitarity V V^dag = 1 : True
NUM CKM |V| (rows ud us ub / cd cs cb / td ts tb) : [[0.97435, 0.225008, 0.003732], [0.224876,
    ...0.973489, 0.04183], [0.008583, 0.041109, 0.999118]]
NUM Jarlskog J = Im(V_us V_cb V_ub* V_cs*) : 3.1169935287554706e-05
J = c12 c23 c13^2 s12 s23 s13 sin(delta) = 3.12e-5 (PDG 2024: 3.12 +0.13 -0.12 e-5) : True
NUM Wolfenstein lambda = s12, A = s23/s12^2, rhobar + i etabar = -V_ud V_ub*/(V_cd V_cb*) :
    ...(0.22501, 0.8261981634695225, np.float64(0.15918443594247167),
    ...np.float64(0.35227461890586964))
Wolfenstein parameters agree with the PDG fit (A 0.826, rhobar 0.1591, etabar 0.3523) within
    ...their quoted uncertainties : True
Yukawa matrices Y_u = sqrt2 diag(m_u,m_c,m_t)/v, Y_d = sqrt2 V diag(m_d,m_s,m_b)/v: singular
    ...values x v/sqrt2 = masses, |U_uL^dag U_dL| = |V| : True
NUM top Yukawa y_t = sqrt2 m_t/v : 0.9911341438317451
PMNS unitarity : True
NUM PMNS |U| and J_PMNS : ([[0.82328, 0.54796, 0.14816], [0.27491, 0.61298, 0.74073], [0.49663,
    ...0.56921, 0.65526]], np.float64(0.0017359130944756358))
NUM Dirac neutrino masses (lightest m1 = 0, eV) and their sum : (0.0, 0.008654478609367522,
    ...0.05033885179461287, 0.058993330403980394)
neutrino Yukawa matrix Y_nu = sqrt2 U diag(m1,m2,m3)/v: masses and |U| recovered (singular
    ...value decomposition) : True
NUM largest neutrino Yukawa sqrt2 m3/v : 2.891316340688847e-13
vacuum: probabilities P(a -> b) form a doubly stochastic matrix (rows and columns sum to 1) :
    ...True
vacuum: amplitude result = delta_ab - 4 sum_{i>j} Re(X_ij) sin^2 Delta_ij + 2 sum_{i>j}
```

```
    ...Im(X_ij) sin 2Delta_ij, X_ij = U_ai* U_bi U_aj U_bj* : True
NUM DUNE vacuum (L = 1300 km, E = 2.5 GeV): P(mu->e), P(mubar->ebar), P(mu->mu) :
    ...(np.float64(0.04883685296656589), np.float64(0.05019851034574761),
    ...np.float64(0.012813139189347619))
CPT: P(a->b) = P(bbar->abar); CP asymmetry P(mu->e) - P(mubar->ebar) proportional to J
    ...(vanishes for delta = 0, pi) : True
NUM V_CC = sqrt2 G_F n_e for rho Y_e = 1 g/cm^3 [eV] (textbook 7.63e-14 eV) :
    ...7.632466872241316e-14
V_CC(rho Y_e = 1 g/cm^3) = 7.63e-14 eV : True
matter (constant density): unitary evolution, probabilities doubly stochastic : True
NUM DUNE in matter (rho = 2.848 g/cm^3, Y_e = 0.5): P(mu->e), P(mubar->ebar) :
    ...(np.float64(0.07232693782692827), np.float64(0.029191799469459462))
matter: a flavour-universal potential has no effect (only differences of potentials matter) :
    ...True
two-flavour MSW: sin^2 theta_m from diagonalization = formula; maximal mixing (sin^2 2theta_m =
    ...1) at 2 E V = dm^2 cos 2theta : True
NUM Sun: P_ee (numerical through the solar density profile, adiabatic formula) for E = 0.3, 1,
    ...5, 10 MeV : {0.3: (np.float64(0.5251624291065616), 0.5251645055880894), 1.0:
    ...(np.float64(0.4584114174121251), 0.4584122490661078), 5.0:
    ...(np.float64(0.3103377102301538), 0.3103335111666083), 10.0:
    ...(np.float64(0.2979217906988696), 0.29791748878645763)}
Sun: numerical P_ee = adiabatic MSW formula to 2e-3 at E = 0.3, 1, 5, 10 MeV : True
Sun: P_ee falls from near the vacuum average c13^4 (1 - sin^2(2 theta12)/2) + s13^4 (within
    ...0.03 at 0.3 MeV) to about 0.31 at 10 MeV : True
NUM supernova MSW resonance densities for E = 20 MeV, Y_e = 0.5: H (dm31, theta13), L (dm21,
    ...theta12) [g/cm^3] : (1587.1391520947036, 18.939748107619376)
NUM cosmic neutrino background: T_nu = (4/11)^(1/3) T_gamma [K], number density per flavour
    ...[cm^-3] : (1.9453688391750839, 112.01641316574336)
CNB: T_nu = 1.945 K and n = 112 cm^-3 per flavour : True
brane representation: gamma_mu (x) 1_4 (mu = 0..3) and gamma5 (x) rho_k (k = 4..7) satisfy
    ...{R_A, R_B} = 2 eta_AB : True
intertwiner U (Schur average over the 256 Clifford products): Gamma17[A] U = U R_A for A =
    ...0..7, U unitary : True
8D chirality at the brane: chi17 = U (gamma5 (x) rho4567) U^dag, rho4567 Hermitian with
    ...eigenvalues +1, +1, -1, -1 : True
bilinears at the brane: Psibar Psi' = sum_a psibar_a psi'_a and Psibar Gamma^mu Psi' = sum_a
    ...psibar_a gamma^mu psi'_a (mu = 0..3) : True
Kaluza-Klein: the 16x16 operator gamma.p (x) 1 + gamma5 (x) rho.q/b0 - m is singular exactly on
    ...p0^2 - |p|^2 = m^2 + |q|^2/b0^2 : True
KK operator: X^2 = -|q|^2/b0^2 and X anticommutes with gamma^mu (x) 1 : True
portal mixing: light masses (eV) and heavy masses (GeV) of the 7x7 Dirac mass matrix : ([0.0,
    ...0.008654465673872413, 0.05033886016084743], [1.0, 1.0, 1.00000000025,
    ...1.0000000007499998])
portal mixing: N N^dag = 1 - mu mu^dag/m^2 to second order (non-unitary light mixing), (1 - N
    ...N^dag)_ee : (True, np.float64(9.99999860695766e-10))
non-unitary oscillations: sum_b P(a->b) = (N N^dag)_aa^2-type deficit < 1 (zero-distance
    ...effect), complemented by the heavy states : True
portal decoupling: for mu -> 0 the light block becomes the PMNS matrix and the oscillation
    ...probabilities the three-flavour vacuum ones (section 19) : True
NUM vector portal (V_P,e = 0.2 V_CC): P(mu->e) at DUNE with and without the dirac16complex
    ...potential : (np.float64(0.07704998293140915), np.float64(0.07232693782692827))
NUM age of the universe from the brane Friedmann equation (Planck 2018: H0 = 67.4, Omega_m =
    ...0.315) [Gyr] : 13.790728126409226
age 13.79 Gyr (Planck 2018: 13.787 +- 0.020) : True
ADD relation: 3 Ha^2 = kappa8 rho4/V_perp = 8 pi G4 rho4 with G4 = G8/V_perp : True
NUM ADD example (reduced 8D Planck mass 10 TeV): V_perp [GeV^-4], V_perp^(1/4) [m], lowest KK
    ...mass 2 pi/V_perp^(1/4) [MeV] : (5929225000000.0, 3.079186208511254e-13,
    ...4.026524867292262)
boolean checks: 32   failed: 0
VERDICT: ALL CHECKS PASSED
```

4.4 Hashes of the files used to build 7+1.nb and its documentation

file	SHA-256
7+1.nb	060942488caf4f45120ffd37be0123e96992a31cd7fdbde09ee4dc4ced4f5d3a
4+4.nb	740023233eaa0cd24ca0e2e7010094671b1d314b698114c08db94050d3e0aed5
tools/cayley_dickson_formulas.py	bdde8c4c8199b09ef36ae4e15d5d53bc62744fc85b4c5ba52eab5516f77b5d5f
tools/cayley_dickson_match.py	73f2cc584bbe1a3620c50890ffe5fb299857b04a14bc34fed2f9eaefd2e36c4e
tools/classify_clifford.py	42291276f46b046183fdbb7b62be7c5cedfd695fe9da96ac256afa23580930a9
tools/doc_common.py	eb63a44f3da35fc75dae8420186867f85fecfa15415eefa108b5b0f23c646385
tools/doc_dirac16.py	7916c204596a471015e9408616aaca1b7b89575adc774b808c468f337e8418de
tools/doc_dirac16_common.py	9324b4fae59e51019125ebe09157f3c8c9489560e1383a697fe6c22a3a93e208
tools/doc_dirac16_provenance.py	9ac843eac503c8f3d1417a67e82765c0f896422171a897a3200b67f80d420dfe
tools/doc_dirac16_symbols.py	70cee4b84501c0f7a7a02105e39acfa6e21c61835a667eca28e43e6ad65cf674
tools/doc_gravity.py	6d0da11131b4f0473855658cdb1c2a34afbdac630cfcf528a9ecbe5cf5ef3d90
tools/doc_gravity_provenance.py	2809bf6cc4f6c44c4f9a66422321329103bb4d231cbc3c1e7eea794026fe77f4
tools/doc_gravity_symbols.py	da852b5feb4a7d68b44da53eb254471d1c8546d305b6dda3c8e59440e1372ae8
tools/doc_ks_gravity.py	e62dffaadee563428788e9600ba14c92acc346186d58d70a9b41193e917f9ff8
tools/doc_ks_gravity_provenance.py	1a97d4be904ff2921c426ae92dee4ef3a3eb9dde20f2f9ef55babb5a9923e4f1
tools/doc_ks_gravity_symbols.py	afa9b5b30b13e5ab90a2cf74bd6ae5ef8a5dbaa34c25ce33a1ce6011033f9fbd
tools/doc_provenance.py	49e7b60c6ce31ad7869b3b9c3eaefd28517e8b4ffd7e4d8c0e9651d461bb56a1
tools/doc_readme.py	1b99079a18c39f9ba7a73465982787c9b9139079d3a4af98e65a261bb5eaacff
tools/doc_readme_stage3.py	8076bbaf4f0c8854a9411f95cd1637fc1dcfeae1fc45f2701d88ff7f356a67d6
tools/doc_sm.py	faa244fed5ec09465f576661ad601def9e815e56389361e037e98fd5b45553c0
tools/doc_sm_provenance.py	ab0f9d1cce27f2250bfa278ff72670bd40339cfa8ca0ecc025c3e099a2d4118a
tools/doc_sm_symbols.py	353df4989623fed65925777ce46759ff296f3cec3bd42797123b7b4b0b3d9766
tools/doc_stage3_common.py	2364ce6eff9533051a9b56a5435e59039cf9417c03a477d75d40f68c8dd4b1d6
tools/doc_symbols.py	f5bf59d7962459f7e8bc06581852e0be3b695f7cec7513b01a9e91fcfa1fa507
tools/docgen.py	5db6f40cb1eb10e48da5b69c3893829737c3d8d2e90edcb28311a980c1aa5fea
tools/eval_cells_log.wl	833b6bd854cab253e5bb935c28f28ab50b286f409d130f433fa5dfb9b9236bbe
tools/extra71.py	9935ef99cd9300f2a7d4f06d36cf7e173912ebae61b6feff448ebe0588853168
tools/git_push.wl	ac05a6d0256834b48cc9fa943fa247bccc574f723e68f053d00bfba9aca2e21a
tools/lovelock71.py	d93ecbd0f82adb972218d59dfe32999a4af944945ca9758607f3a633e1af95a1
tools/make_7p1_nb.py	2fac94dfd0c9fcf3aed8d6e90a77fbbb8ab1c8450819ff591b7e5cf7b5787d6f
tools/make_docs.py	2c71a87dc93d5612106129a671af4f67b351d4f290e586388f42106a39b74f81
tools/nb_part_gravity.py	55d16f735e16df7a7cd2aa70d322969bc6cf82c777965f112b5a9acf46a6ad5f
tools/nb_part_ks_gravity.py	886814fedc4cd015c63b950ee83724025ba21fd6c3e56b68dcb7f2f07afdc205
tools/nb_part_sm.py	20782f1fe557fd3f3dea7417e89ecd42a1602f7a48599fcdcfd0cff77f120ecc
tools/nbparse.py	ad2255a8864be7447a55023840b327f789cb8c9d868d42261b19261653847423
tools/nbwrite.py	dcbe8ce5b3203c89a1d17fbe946e1b6b7bb7f8895afa944743d651cfefadeffb
tools/octalg.py	36b50120b69a963e252e07d60ec007755220822b4ad339716ea7aed93d58eb24
tools/ref44.py	ab100c3ebcc055fbe59c132a02f80ecb17630e934e99d59f88e919ec8064faf5
tools/ref71.py	4ac783b1ab221317387f2a74e0be60f114fd64940fe3ea96af79b33ee7dcb8b0
tools/ref_dirac16.py	a6e139ebe5199621a95c8adc772bc3e1e9b452c0ca20a963c536cbfc6539319c
tools/ref_gravity.py	2bfdc087ed4dc7d84258e04449d11141b415f54b992e65d42fa42cace08ff6bd
tools/ref_ks_gravity.py	8d082985c5161254d7c6294c0310f7807aa037bff27a8cd3ec4d3f8a9f6e0550
tools/ref_sm.py	d36d23f472c8339d80f1d0b1eb46c509398562053ead7033d96b054782e73b49
tools/run_eval.wl	85ba54f18a8362a69c991e66a207230dd001ba18e52639fc1edf7cac761aac30
tools/verify_7p1_nb.py	d3aaeeeb7f380f63d4e7610c5b4235973cadd1d6db0ebf09ecbd65d561795d17
tools/watch_requests.wl	d9ceedb0b45ac3cf9b4dd8519ac164953f2c80e5188b1557fe20ef5107ae352d

5. Process and environment

- Workspace: `D:\Developer\github\7+1` on Windows; Python and LaTeX ran in a Linux shell with that folder mounted. Intermediate files go to `build/`, which the repository's `.gitignore` excludes.
- The notebook is generated by `tools/make_7p1_nb.py` (cells written with `tools/nbwrite.py`), evaluated in place by `tools/run_eval.wl` in Wolfram 15.0.1, and cross-checked by `tools/verify_7p1_nb.py`.
- Python 3.13.17 with sympy 1.14.0 and numpy 2.5.3 for the documentation run of 2026-10-27 (exact rational arithmetic for every identity; the first stage was checked with Python 3.10.12, sympy 1.14.0 and numpy 2.2.6).

- The documentation is generated by `tools/make_docs.py` from the modules `docgen.py`, `doc_common.py`, `doc_dirac16_common.py`, `doc_readme.py`, `doc_symbols.py` and `doc_provenance.py` of the folder tools/ (and the number formatting of `doc_dirac16.py`): one source renders Markdown and LaTeX, and every matrix, table and check list is computed or parsed, never typed. The LaTeX is compiled with XeLaTeX (TeX Live 2022, `latexmk -xelatex`, fonts Latin Modern and DejaVu Sans Mono).
- History: the first algebraic experiments were made in a temporary cloud sandbox and then moved into `tools/` of the workspace, where everything was re-run. A Wolfram 15.0 kernel window was opened once by mistake before Wolfram 15.0.1 was available; no result comes from it. Only Wolfram 15.0.1 evaluated 7+1.nb.
- Corrections made during the review: an earlier draft called the real on-shell amplitudes of notebook section 5.4 "Majorana spinors"; that label was removed, and section 5.5 was added to show which real spinor FIELDS exist (massless with 16 components, massive only with the 32×32 matrices).
- The evaluated notebook was rendered section by section in the Wolfram 15.0.1 front end (`Rasterize`) and inspected. Characters that the front end's text font shows badly are written differently in text cells: the double-struck O and the arrow of $\vec{x}$ as small inline formulas, the script ℓ in the font Cambria Math, and $\not{p}$, $\not{k}$ as `pslash`, `kslash`.
- Second stage (sections 7–10 of 7+1.nb): all Wolfram Language work ran in the front end `C:\Program Files\Wolfram Research\Wolfram\15.0.1\WolframNB.exe`. Jobs were requested without keyboard input through `tools/watch_requests.wl` (loaded once in that front end; every 10 seconds it reads `build/request.txt` and evaluates `tools/run_eval.wl`, `tools/git_push.wl` or a development file); new cells were developed with `tools/eval_cells_log.wl`, which evaluates a generated notebook expression by expression, logs messages and checks, and stops at the first failure. `tools/run_eval.wl` evaluates with a time limit of one hour. Pushes run the local Git for Windows from the front end (`tools/git_push.wl`, no login prompt) to `https://github.com/once-ere/7plus1.git`, the new name of the repository.
- Second stage, Python: `tools/ref_dirac16.py` (sympy, mpmath, numpy) is the independent reference; `tools/verify_7p1_nb.py` was extended to sections 7–9 and to the comparison of sections 0–6 with the backup of the first stage.
- Third stage (sections 11–20): the new sections were first evaluated cell by cell in a fresh parallel kernel of the same front end (a development file runs the input cells of sections 0–6, 7.1, 7.10 and 11–19 and writes a progress line around every cell and the list of failed checks); this exposed a name clash (a helper named like the zero vector `zero8` of section 1, renamed `asMat8`) and a formulation of the exponential solutions that did not finish (xAct with symbolic exponentials; replaced by the substitution into the xAct components of the cosmological metric). The complete notebook was then evaluated by `tools/run_eval.wl` as before.
- Third stage, Python: `tools/lovelock71.py` (curvature and pair-set Lovelock engine), `tools/ref_gravity.py`, `tools/ref_ks_gravity.py`, `tools/ref_sm.py` (sympy, numpy, scipy, mpmath) are the independent references; the documentation of the third stage is generated by `tools/doc_stage3_common.py`, `tools/doc_gravity*.py`, `tools/doc_ks_gravity*.py`, `tools/doc_sm*.py` and `tools/doc_readme_stage3.py`.

6. The second stage: sections 7–10 of 7+1.nb

This section records the provenance of the Dirac equation and the Kohn–Sham theories of the field dirac16complex (sections 7–10 of 7+1.nb). Notation: $\Gamma_A = \Gamma_A^{(1,7)}$, $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A$, $\Gamma^0 = \Gamma_0$, $\chi = \chi^{(1,7)} = i\Gamma_8^{(1,7)}$, Ψ the field dirac16complex, $\bar{\Psi} = \Psi^\dagger\Gamma^0$, $\vec{x} = (x_1, \ldots, x_7)$, $\vec{p} = (p^1, \ldots, p^7)$.

6.1 The task

1. Back up 7+1.nb and append to it; the first part stays unchanged.
2. A checked Dirac equation on the real 7+1 spacetime for a 16-component quantized fermion field named dirac16complex whose components are complex-valued Grassmann operators, with eight PURELY REAL anticommuting 16×16 Dirac matrices taken from 7+1.nb.
3. Chiral projections writing the 16-component equation as two 8-component equations, coupled when the mass is not zero.
4. A 16-component Kohn–Sham theory of the field coupled to itself, and a chiral version using the chiral projections.
5. Complete, checked documentation (every document with a provenance file and a symbol file, Markdown and LaTeX, compiled to PDF, each self-contained); every completed stage pushed to github.com/once-ere/7plus1 (the repository was renamed from 7+1 to 7plus1); only the Wolfram front end 15.0.1 is used.

6.2 The first part of the notebook is unchanged

The evaluated notebook of the first stage was copied, in the workspace, to the file below (the folder backups/ is excluded from the repository by its .gitignore). `tools/verify_7p1_nb.py` compares every input, text and section cell of sections 0–6 of the new 7+1.nb with that copy: all are identical.

```
backups/7+1_2026-10-26_before_dirac16complex.nb
306963 bytes
SHA-256 f7abe7279b5540e914de876dc055c199bd491b9c2d9522bf28dbc23689e237ce
```

6.3 Derivations and decisions

Dirac matrices. The real generators $\Gamma_A^{(1,7)}$ of section 2.5 are used unchanged: $\Gamma_0 = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i = \left(\begin{smallmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{smallmatrix}\right)$, $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}^{(1,7)} 1_{16}$. The opposite sign convention $\eta^{(7,1)}$ has no real 16×16 realization (section 2.7), so this is the only real choice. **The factor i.** With real Γ_A, $(\Gamma^A\partial_A)^2 = \frac{1}{2}\{\Gamma^A,\Gamma^B\}\partial_A\partial_B = \partial_0^2 - \partial_1^2 - \cdots - \partial_7^2$, hence

$$\big(i\Gamma^A\partial_A + m\big)\big(i\Gamma^A\partial_A - m\big) = -\big(\partial_0^2 - \partial_1^2 - \cdots - \partial_7^2 + m^2\big)1_{16},$$
$$\Gamma^0\big(i\Gamma^A\partial_A - m\big) = i\partial_0 - H_{\mathrm{D}}, \qquad H_{\mathrm{D}} = -i\Gamma^0\Gamma^i\partial_i + m\Gamma^0.$$

Real amplitudes. For $e^{-i(Ex_0 - \vec{p}\cdot\vec{x})}$, with $P_\tau = p^i\tau_i^O$ ($P_\tau^{\mathsf{T}} = -P_\tau$, $P_\tau^2 = -\vec{p}^{\,2}1_8$):

$$H_{\mathrm{D}}(\vec{p}) = \begin{pmatrix} 0 & m1_8 + P_\tau \\ m1_8 - P_\tau & 0 \end{pmatrix}, \qquad (m1_8 + P_\tau)(m1_8 - P_\tau) = (m^2 + \vec{p}^{\,2})1_8 = E^2 1_8,$$
$$H_{\mathrm{D}}(\vec{p}) \begin{pmatrix} E\,e_s \\ (m1_8 - P_\tau)e_s \end{pmatrix} = \begin{pmatrix} E^2 e_s \\ E(m1_8 - P_\tau)e_s \end{pmatrix} = E \begin{pmatrix} E\,e_s \\ (m1_8 - P_\tau)e_s \end{pmatrix},$$
$$w_s = \frac{1}{\sqrt{E}} \begin{pmatrix} E\,e_s \\ (m1_8 - P_\tau)e_s \end{pmatrix}, \qquad w_s^{\mathsf{T}} w_r = 2E\,\delta_{sr}.$$

Chirality. The real matrices anticommuting with every Γ_A are $c\,\Gamma_8^{(1,7)}$ (section 2.9), with square $-c^2 1_{16}$; the chirality is therefore the complex $\chi = i\Gamma_8^{(1,7)} = \left(\begin{smallmatrix} 0 & -i1_8 \\ i1_8 & 0 \end{smallmatrix}\right)$. With $E_\pm$ the 16×8 matrices $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1_8 \\ -i1_8 \end{smallmatrix}\right)$ and $\frac{1}{\sqrt{2}}\left(\begin{smallmatrix} -i1_8 \\ 1_8 \end{smallmatrix}\right)$ ($\chi E_- = -E_-$, $\chi E_+ = E_+$, $E_-E_-^\dagger = P_L^{(1,7)}$, $E_+E_+^\dagger = P_R^{(1,7)}$), $\Psi_L = E_-^\dagger\Psi$, $\Psi_R = E_+^\dagger\Psi$ and

$$E_-^\dagger\Gamma^0 E_+ = E_+^\dagger\Gamma^0 E_- = 1_8, \qquad E_\mp^\dagger\Gamma^0 E_\mp = 0, \qquad E_\mp^\dagger\Gamma^0\Gamma^i E_\mp = \mp i\tau_i^O, \qquad E_-^\dagger\Gamma^0\Gamma^i E_+ = 0,$$

$$E_-^\dagger\Gamma^0\big(i\Gamma^A\partial_A-m\big)\Psi=\big(i\partial_0+\tau_i^O\partial_i\big)\Psi_L-m\Psi_R,\qquad E_+^\dagger\Gamma^0\big(i\Gamma^A\partial_A-m\big)\Psi=\big(i\partial_0-\tau_i^O\partial_i\big)\Psi_R-m\Psi_L.$$

Self-coupling. The four Lorentz-invariant contact interactions without derivatives used in relativistic point-coupling models (scalar, pseudoscalar, vector, axial vector) with real couplings, and real sources that define the densities:

$$\begin{aligned}\mathcal{L}=\bar\Psi\big(i\Gamma^A\partial_A-m\big)\Psi-\bar\Psi\big(\mathsf{s}+i\chi\mathsf{p}+\Gamma^A(\mathsf{v}_A+\chi\mathsf{a}_A)\big)\Psi\\-\frac{g_S}{2}(\bar\Psi\Psi)^2-\frac{g_P}{2}(\bar\Psi i\chi\Psi)^2-\frac{g_V}{2}(\bar\Psi\Gamma^A\Psi)(\bar\Psi\Gamma_A\Psi)-\frac{g_A}{2}(\bar\Psi\Gamma^A\chi\Psi)(\bar\Psi\Gamma_A\chi\Psi).\end{aligned}$$

The tensor and higher-rank channels are not included. With $\hbar=c=1$ every g has mass dimension -6 (an effective theory: no-sea approximation, or a momentum cutoff for the sea). **Hohenberg–Kohn.** For $H[w]=H_0+H_{\rm int}+\int w\cdot n$ with $w\cdot n=\mathsf{s}\rho_S+\mathsf{p}\rho_P+\mathsf{v}_Aj^A+\mathsf{a}_Aj_5^A$, two sources $w\neq w'$ with different ground states and the same densities would give $E<E'+\int(w-w')\cdot n$ and $E'<E+\int(w'-w)\cdot n$, so $E+E'<E+E'$; hence $E[w]=\min_n(F[n]+\int w\cdot n)$. **Kohn–Sham.** Varying $T_s+\int w\cdot n+E_H+E_{xc}$ with respect to $\Phi_k^\dagger$ under $\int\Phi_k^\dagger\Phi_l=\delta_{kl}$ (multipliers λ_k), with $\delta\rho_S/\delta\Phi^\dagger=\Gamma^0\Phi$, $\delta\rho_P/\delta\Phi^\dagger=\Gamma^0i\chi\Phi$, $\delta j^A/\delta\Phi^\dagger=\Gamma^0\Gamma^A\Phi$, $\delta j_5^A/\delta\Phi^\dagger=\Gamma^0\Gamma^A\chi\Phi$, gives $H_{\rm KS}\Phi_k=\lambda_k\Phi_k$ with

$$\begin{aligned}H_{\rm KS}&=\Gamma^0\Gamma^i(-i\partial_i+\mathcal{V}_i)+\Gamma^0(m+\mathcal{S})+\mathcal{V}_0+\Gamma^0i\chi\mathcal{P}+\chi\mathcal{A}_0+\Gamma^0\Gamma^i\chi\mathcal{A}_i,\\ \mathcal{S}&=\mathsf{s}+g_S\rho_S+\frac{\delta E_{xc}}{\delta\rho_S},\qquad \mathcal{P}=\mathsf{p}+g_P\rho_P+\frac{\delta E_{xc}}{\delta\rho_P},\\ \mathcal{V}_A&=\mathsf{v}_A+g_Vj_A+\frac{\delta E_{xc}}{\delta j^A},\qquad \mathcal{A}_A=\mathsf{a}_A+g_Aj_{5A}+\frac{\delta E_{xc}}{\delta j_5^A},\end{aligned}$$

where $E_H=\int[\frac{g_S}{2}\rho_S^2+\frac{g_P}{2}\rho_P^2+\frac{g_V}{2}j^Aj_A+\frac{g_A}{2}j_5^Aj_{5A}]$ and $T_s=\sum_k\int\Phi_k^\dagger H_{\rm D}\Phi_k$. **Decision (no sea):** the orbitals are the occupied positive-energy states. **Exchange.** Wick's theorem for a Slater determinant, $\langle c_a^\dagger c_c^\dagger c_dc_b\rangle=\varrho_{ba}\varrho_{dc}-\varrho_{da}\varrho_{bc}$ with $\varrho_{xy}=\sum_k\Phi_{k,x}\Phi_{k,y}^*$ and $\mathcal{R}=\varrho\,\Gamma^0$, gives $\langle:(\bar\Psi O_1\Psi)(\bar\Psi O_2\Psi):\rangle=\mathrm{tr}(O_1\mathcal{R})\mathrm{tr}(O_2\mathcal{R})-\mathrm{tr}(O_1\mathcal{R}O_2\mathcal{R})$, hence the exact exchange energy

$$E_x=-\frac{g_S}{2}\mathrm{tr}(\mathcal{R}\mathcal{R})-\frac{g_P}{2}\mathrm{tr}(i\chi\mathcal{R}\,i\chi\mathcal{R})-\frac{g_V}{2}\mathrm{tr}(\Gamma^A\mathcal{R}\Gamma_A\mathcal{R})-\frac{g_A}{2}\mathrm{tr}(\Gamma^A\chi\mathcal{R}\Gamma_A\chi\mathcal{R}).$$

Chiral Kohn–Sham. With $\Phi_{kL}=E_-^\dagger\Phi_k$, $\Phi_{kR}=E_+^\dagger\Phi_k$ and the blocks above, $\kappa=\sum_k\Phi_{kL}^\dagger\Phi_{kR}$, $\rho_S=\kappa+\kappa^*$, $\rho_P=i(\kappa-\kappa^*)$, $\mathcal{M}=m+\mathcal{S}+i\mathcal{P}$, $\mathcal{V}_A^{L,R}=\mathcal{V}_A\mp\mathcal{A}_A$:

$$\begin{aligned}\big[\mathcal{V}_0^L-\tau_i^O\partial_i-i\tau_i^O\mathcal{V}_i^L\big]\Phi_L+\mathcal{M}\Phi_R&=\lambda\Phi_L,\\ \mathcal{M}^*\Phi_L+\big[\mathcal{V}_0^R+\tau_i^O\partial_i+i\tau_i^O\mathcal{V}_i^R\big]\Phi_R&=\lambda\Phi_R,\\ e^{i\vartheta\chi}&=\cos\vartheta\,1_{16}-\sin\vartheta\,\Gamma_8^{(1,7)}:\qquad\kappa\to e^{2i\vartheta}\kappa,\\ \Delta E_H&=\frac{g_S-g_P}{2}\big[(e^{4i\vartheta}-1)\kappa^2+(e^{-4i\vartheta}-1)\kappa^{*2}\big].\end{aligned}$$

Uniform matter. Units $m=1$; all 8 spin states filled to k_F; $|S^6|=16\pi^3/15$ is the area of the unit sphere in $\mathbb{R}^7$ and $c_7=8|S^6|/(2\pi)^7=1/(15\pi^4)$. With the energy density $\mathcal{E}$, its kinetic part $\mathcal{E}_{\rm kin}$, ρ_S and the density ρ:

$$\begin{aligned}\rho&=\frac{k_F^7}{105\pi^4},\qquad\rho_S=c_7\int_0^{k_F}\frac{k^6m^*\,dk}{\sqrt{k^2+m^{*2}}},\qquad\mathcal{E}_{\rm kin}=c_7\int_0^{k_F}k^6\sqrt{k^2+m^{*2}}\,dk,\\ m^*&=m+g_S\rho_S(m^*),\qquad\mathcal{E}=\mathcal{E}_{\rm kin}-\frac{(m^*-m)^2}{2g_S}+\frac{g_V}{2}\rho^2,\\ E/A&=\frac{\mathcal{E}}{\rho},\qquad E_F=g_V\rho+\sqrt{k_F^2+m^{*2}}.\end{aligned}$$

Because $\partial\mathcal{E}/\partial m^*=0$ is the gap equation, $d\mathcal{E}/d\rho=E_F$ at every density, and $\frac{d}{dk_F}\frac{\mathcal{E}}{\rho}=\frac{c_7k_F^6}{\rho}(E_F-\frac{\mathcal{E}}{\rho})$ vanishes at saturation: $E/A=E_F$ (Hugenholtz–Van Hove). **Decisions:** $g_S=-10^5$, $g_V=10^4$, $g_P=g_A=0$ (binding and saturation); root finding by Brent's method with 50-digit (gap equation) and 40-digit (saturation) working precision. **Dirac sea.** With a cutoff $\Lambda_{\rm c}$, $g_S=g_P=g$ and a uniform $\mathcal{M}$:

$$\begin{aligned}J(M)&=\frac{1}{120\pi^4}\int_0^{\Lambda_{\rm c}}\frac{q^6dq}{\sqrt{q^2+M^2}},\qquad K(M)=\frac{1}{120\pi^4}\int_0^{\Lambda_{\rm c}}q^6\sqrt{q^2+M^2}\,dq,\\ \kappa&=-4\mathcal{M}^*J(|\mathcal{M}|),\qquad\mathcal{M}=m-8gJ(|\mathcal{M}|)\mathcal{M},\qquad g_c=-\frac{1}{8J(0)}=-\frac{90\pi^4}{\Lambda_{\rm c}^6},\\ \mathcal{E}_{\rm vac}&=-8K+8J\,\mathrm{Re}[(\mathcal{M}-m)\mathcal{M}^*]+32g|\mathcal{M}|^2J^2.\end{aligned}$$

Decisions: $\Lambda_{\rm c}=1$, $g=\frac{6}{5}g_c$, and $m=\frac{1}{100}$ for the explicit-breaking example.

6.4 Literature used in the second stage

- P. A. M. Dirac, Proc. R. Soc. Lond. A 117 (1928) 610 — $(i\gamma^\mu\partial_\mu - m)\psi = 0$, $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}$.
- P. Hohenberg and W. Kohn, Phys. Rev. 136 (1964) B864 — $E[v] = \min_n(F[n] + \int vn)$; W. Kohn and L. J. Sham, Phys. Rev. 140 (1965) A1133 — $[-\frac{1}{2}\nabla^2 + v + v_H + v_{xc}]\varphi_k = \varepsilon_k\varphi_k$, $n = \sum_k |\varphi_k|^2$ (atomic units).
- A. K. Rajagopal and J. Callaway, Phys. Rev. B 7 (1973) 1912; R. M. Dreizler and E. K. U. Gross, *Density Functional Theory*, Springer (1990) — the energy as a functional of the current j^A, the Dirac form of the Kohn–Sham equations, the no-sea approximation.
- G. C. Wick, Phys. Rev. 80 (1950) 268; M. Fierz, Z. Phys. 104 (1937) 553 — contractions in a Slater determinant (above), and $\sum_K(\Gamma_K)_{ab}(\Gamma_K^{-1})_{cd} = 16\delta_{ad}\delta_{cb}$ for the 256 products Γ_K of the generators.
- J. D. Walecka, Ann. Phys. 83 (1974) 491 — mean-field saturation of uniform matter, $m^* = m - \frac{g_\sigma^2}{m_\sigma^2}\rho_S$, $V_0 = \frac{g_\omega^2}{m_\omega^2}\rho$ (here $g_S = -g_\sigma^2/m_\sigma^2$, $g_V = g_\omega^2/m_\omega^2$); N. M. Hugenholtz and L. Van Hove, Physica 24 (1958) 363 — $E/A = E_F$ at saturation.
- Y. Nambu and G. Jona-Lasinio, Phys. Rev. 122 (1961) 345 — $M = m - 2G\langle\bar\psi\psi\rangle$ with a cutoff and a critical coupling (here $G = -g/2$).

6.5 Verification of the second stage

Sections 7–9 of 7+1.nb contain 74 checks, all True; they are rows of section 4.2 above, and the evaluation is the one of section 4.1. `tools/ref_dirac16.py` recomputes the constructions independently with 90 boolean checks, all True, and its output is printed in section 4.3, together with that of `tools/verify_7p1_nb.py`, which compares the printed spinors, the closed-form integrals and 23 printed numbers with it (at least 17 digits) and the first part of the notebook with the backup.

7. The third stage: sections 11–20 of 7+1.nb

This section records the provenance of the gravity, Kohn–Sham and Standard Model sections (11–19) and of the final summary (20). Notation: $\Gamma_A = \Gamma_A^{(1,7)}$, $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A$, $\eta = \mathrm{diag}(+1,-1,\ldots,-1)$, Ψ the field dirac16complex.

7.1 The task

1. Rebuild the ideas of github.com/once-ere/Dirac_claude on the 7+1 spacetime of this project (one time coordinate, seven space coordinates; never the 4+4 metric of Dirac_claude): the Einstein and Lovelock tensors; generalized frame matrices; a canonical spin connection, with a vanishing total covariant derivative, that couples gravity to dirac16complex; the Kohn–Sham theory of dirac16complex in gravitational fields in the primordial (near $x_0 = 0$), modern and far-future epochs; a generalized self-coupling; and the totality of the Standard Model.
2. Compute the Lovelock tensors as in the author's corrected reference notebook Einstein-Lovelock-7+1-Nash.nb, and do not use its section "FAILED Model metric g71; DO NOT USE!".
3. Document every stage (documentation, provenance and symbol files; Markdown, LaTeX, PDF; self-contained) and push every completed stage to github.com/once-ere/7plus1.

7.2 Inputs and how they were used

The reference notebook Einstein-Lovelock-7+1-Nash.nb (P. L. Nash, 2022, GPL; the author's corrected version) supplied: the xAct packages xCoba and xTras; the manifold with abstract indices a, b, c, d, f, i, j, k, l, m, n, p, q, r, s, y; the chart with coordinate scalars `x0[]`, …, `x7[]`; `MyArrayComponents`; the generalized delta `kδ`; the three Lovelock tensors with their normalization

$$\mathrm{Lovelock}_k = -\frac{1}{2^{k+1}}\, R_{c_1 d_1}{}^{f_1 i_1} \cdots R_{c_k d_k}{}^{f_k i_k}\, \delta^{b\, c_1 d_1 \cdots c_k d_k}_{a\, f_1 i_1 \cdots f_k i_k} \quad (k = 1, 2, 3);$$

the evaluation of components by `CD = LC[g]`; and the vacuum field equations $0 = -\Lambda\delta + H^{-2} w_1 \mathrm{Lovelock}_1 + H^{-4} w_2 \mathrm{Lovelock}_2 + H^{-6} w_3 \mathrm{Lovelock}_3$ with the procedure of solving them for w_2, w_3. The model metric of its FAILED section was not used anywhere. The only change: the signature declaration $\{7, 1, 0\}$ became $\{1, 7, 0\}$ to match η of this notebook (it does not enter the Lovelock tensors with mixed indices, and `SignDetOfMetric` is -1 for both declarations, as checked with xTensor 1.2.0). xAct versions as printed by the notebook: xPerm 1.2.3, xTensor 1.2.0, xCoba 0.8.6, xTras 1.4.2.

```
Einstein-Lovelock-7+1-Nash.nb   SHA-256 b2dcb7b9f435d644da085fdd9e738a0cc80d3268ef28606ff9f367ed
    ...6828fdfc
Einstein-Lovelock-7+1-Nash.pdf  SHA-256 6580f9fcaff9c7f1337df35a7fce6b4b04c022d0e2d68954e8b3954b
    ...30f45125
```

Data of the Standard Model part: Particle Data Group 2024/2025 (G_F, boson masses, quark masses, CKM parameters), NuFit-6.0 (PMNS parameters, normal ordering, without Super-Kamiokande atmospheric data), Planck 2018 (cosmological parameters).

7.3 Decisions

#	decision	reason
1	Christoffel symbols written $\{^{\lambda}_{\mu\nu}\}$; Landau–Lifshitz curvature	Γ is reserved for Dirac matrices; with $(+,-,\ldots,-)$, $G^{\mu}{}_{\nu} - \bar{\Lambda}\delta^{\mu}_{\nu} = \kappa T^{\mu}{}_{\nu}$, $T^{0}{}_{0} > 0$
2	the Lovelock tensors are computed twice (xAct; pair-set sums)	exact agreement is an independent check of the reference method
3	the exponential solutions are obtained from the xAct components of the cosmological metric	a new xAct run with symbolic exponentials does not finish in reasonable time
4	the canonical (Levi-Civita) spin connection, $\Omega_{\mu} = \frac{1}{4}\omega_{\mu AB}\Gamma^{A}\Gamma^{B}$	it is the connection with $\nabla_{\mu}\Gamma_{\nu} = 0$
5	the self-coupling is organized by the nine ranks of the Clifford algebra	each rank is one Lorentz-invariant class; the Fierz matrix closes on them
6	Kohn–Sham matter: homogeneous, spin-saturated, no Dirac sea, adiabatic	the cosmological spacetime is homogeneous; the expansion rates are small compared with the orbital energies
7	the Standard Model on a 3+1 brane of a torus of the auxiliary coordinates	the 7+1 analogue of the fold of Dirac_claude
8	portals are invented local couplings at the brane	the only contact of dirac16complex with the Standard Model besides gravity

7.4 Literature

- D. Lovelock, J. Math. Phys. 12 (1971) 498; D. Lovelock and H. Rund, *Tensors, Differential Forms, and Variational Principles*, Dover (1989), Remark 2 after Eq. (4.38) — the Lovelock tensors (section 12.1).
- J. M. Martín-García et al., xAct: efficient tensor computer algebra for the Wolfram Language (xPerm, xTensor, xCoba), GPL, www.xact.es; T. Nutma, Comput. Phys. Commun. 185 (2014) 1719 (xTras) — the abstract-index computation of section 12.1.
- L. D. Landau and E. M. Lifshitz, *The Classical Theory of Fields*, Pergamon — curvature conventions; E. Kasner, Am. J. Math. 43 (1921) 217 — power-law solutions; G. W. Gibbons and S. W. Hawking, Phys. Rev. D 15 (1977) 2738 — the de Sitter temperature $H/2\pi$.

- S. Navas et al. (Particle Data Group), Phys. Rev. D 110 (2024) 030001 and the 2025 update; I. Esteban et al., JHEP 12 (2024) 216 (NuFit-6.0); Planck Collaboration, Astron. Astrophys. 641 (2020) A6 — the data of section 12.5–12.6.
- N. Arkani-Hamed, S. Dimopoulos, G. Dvali, Phys. Lett. B 429 (1998) 263; L. Wolfenstein, Phys. Rev. D 17 (1978) 2369; S. P. Mikheyev and A. Yu. Smirnov, Sov. J. Nucl. Phys. 42 (1985) 913 — extra dimensions, the MSW effect (section 12.5–12.6).

7.5 Verification of the third stage

Sections 11–19 of 7+1.nb contain 144 checks (gravity 97, Kohn–Sham 24, Standard Model 23), all True; they are rows of section 4.2 above. The independent programs recompute everything: `tools/ref_gravity.py` (99 boolean checks, 99 True), `tools/ref_ks_gravity.py` (29, 29), `tools/ref_sm.py` (32, 32); their complete outputs are printed in section 4.3.

8. Authorship

Notebook, tools and documentation written by Claude (Anthropic, model claude-opus-5-5) for Nash: the first stage (sections 0–6 of 7+1.nb) from the task statement and the notebook 4+4.nb described in section 1, the second stage (sections 7–10) from the task of section 6.1, the third stage (sections 11–20) from the task of section 7.1 and the reference notebook Einstein-Lovelock-7+1-Nash.nb of its author; this document was generated on 2026-10-27.